MILITARY SPECIFICATION

RADIO SET, AN/ARC-116( )

1. SCOPE

1.1 This specification covers one type of amplitude modulation (AM) airborne radio set which operates on any one of 3500 discrete channels spaced at 50 kHz throughout the frequency range of 225.00 through 399.95 MHz and is designated as Radio Set AN/ARC-116( ). (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed A and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced
QQ-S-781	Steel Strapping, Flat

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bag and Sleeve, Interior Packaging
MIL-B-131	Barrier Material, Water Vaporproofed, Flexible

FSC 5821

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MIL-T-713	Twine, Impregnated, Lacing and Tying
MIL-D-3464	Desiccants, Activated, Bagged, Packaging, Use and Static Dehumidification
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-8835	Indicator, Humidity, Card, Chemically Impregnated
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-19500	Semiconductor Device, General Specification For
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, or Pallet-Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization Of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods

DRAWING

USAECOM

SC-DL-595577	Radio Set AN/ARC-116( )

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Description. - Radio Set AN/ARC-116( ) is a lightweight multichannel airborne radio communication set which provides transmission, reception, and retransmission of amplitude modulated (AM) radio signals between air-to-ground and air-to-air on 3500 discrete channels, each spaced 50 kHz over the frequency range of 225.00 through 399.95 MHz. Receiver self test functions, an independent fixed channel guard receiver in the 238.00 through 248.00 MHz frequency range and an emergency transmitter operating at 243.00 MHz are provided. This radio set is also capable of direction finding and X-Mode operation, when interfaced with associated equipment. The radio set operates from an aircraft power supply having a nominal voltage of 27.5 volts dc.

3.2 Construction. - The units on order shall be constructed in accordance with the following equipment drawing and data list:

SC-DL-595577 Radio Set AN/ARC-116( )

3.3 First Article Samples. - The contractor shall furnish the specified number of First Article samples on order, as required by the invitation for bids and contract. (See 4.3 and 6.2)

3.4. Cleaning. -

3.4.1 Parts. - Prior to the application of conformal coating, on those parts affected, and after fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Post cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, and other parts.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Weight (See 4.21). - The weight of the completely assembled unit shall not exceed 8.6 lbs.

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3.6 Controls. - All moveable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.7 Finish, protective (See 4.4). - The units shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc. and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be Film B in accordance with MIL-F-14072.

3.8 Securing of parts. -

3.8.1 General. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.8.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.9 Marking (See 4.4). -

3.9.1 General. - Marking shall conform to MIL-M-13231. Front panel marking shall be in accordance with Group I of this specification and the equipment drawings.

3.9.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the unit(s).

3.9.3 Serial numbers. - The following unit(s) shall be serial numbered:

Radio Set AN/ARC-116( )

3.10 Soldering. -

3.10.1 Soldering. - Soldering shall conform to Requirement 5 of MIL-STD-454 and the equipment drawings.

3.10.2 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc. may be used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

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3.10.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the unit(s); flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.11 Wiring, cabling, bonds and grounds. -

3.11.1 Cabling. - Insulated wires shall be formed into cables, except where operation of the unit(s) would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine, conforming to MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.11.2 Slack. - Wires and cables shall be as short as practical, except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two (2) replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.11.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.11.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable, (1) during and after subjection of the unit(s) to specified service conditions, or (2) after service or repair of the unit(s) in a normal manner.

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3.11.5 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.11.6 Splicing. - Wires in a continuous run between two (2) terminals shall not be spliced during the wiring operation.

3.11.7 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.11.8 Bonds & Grounds. - Mating surfaces requiring RF gaskets, internal grounds and bonds required to reduce current path returns, RF potentials and shock hazards shall be free of anodic film, non-conductive grease, paint and lacquer.

3.12 Conformal coating (See 4.4). - Except as otherwise specified in the equipment drawings, printed wiring assemblies shall be conformally coated with a coating material which conforms to MIL-I-46058, Type ER (Epoxy) or Type PUR (polyurethane). The coating shall be applied to both sides of the cleaned printed wiring assembly including the component leads. These assemblies shall be cleaned of flux and other contaminants prior to application of coating. Cleaning compounds shall have have no deleterious effects on any part of the printed wiring assembly. The conformal coating shall be compatible with all parts of the printed wiring assembly. The coated assemblies shall show no visible bubbling, blistering, wrinkling, cracking, pinholes, or peeling of the coating material or corrosion of printed conductors. Printed wiring assemblies shall be adequately masked or otherwise protected in such a manner that application of conformal coating does not degrade the electrical performance of the assembly. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Electrical and mechanical mating surfaces such as connector contact points, test points, screw threads, bearing surfaces, etc. shall not be coated. When masking those areas which are not to be coated, and tape is used, the tape shall be an electrical grade. When the printed wiring assembly utilizes glass encased components, such components shall be protected against breakage by the conformal coating by covering the component with thin pliant buffer material, prior to coating, such as polyvinylidene

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fluoride or silicone rubber. This buffer material shall be compatible with the conformal coating material and all parts of the printed wiring assembly. In addition, the buffer material shall be fungus and flame resistant; and shall be clear transparent so markings on the component shall be visible. Conformal coatings shall be clear transparent so markings on the component shall be visible.

3.13 Semiconductors (transistors & diodes) (See 4.4). - These electron devices (semiconductors and diodes) shall conform to MIL-S-19500 and the equipment drawings.

3.14 Tropicalization of materials. - Materials shall be treated as follows:

3.14.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.14.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.14.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.14.4 Statement of treatment. - The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.15 Service conditions (See 4.8). - The unit(s) shall meet the service conditions as specified in MIL-E-5400 for Class 1 equipment and meet the specification requirements, except as specified herein, after a warm-up period not to exceed 60 seconds in the receive mode. The duty cycle shall be five minutes receive and one minute transmit.

3.15.1 Temperature-altitude combination (See 4.8.1). - The unit(s) shall be subjected to and meet the temperature-altitude requirements with a non-operating minimum exposure of 24 hours at -65°F, atmospheric pressure, and 4 hours minimum exposure at 160°F, atmospheric pressure. The unit shall be operable and conforming to limits specified from -25°F through +131°F.

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The unit(s) shall be operable from -65°F to -25°F and no measurement shall degrade by more than 6 db if specified in db or by more than a factor of 2 if the limit is specified in anything but db except that the audio output shall be adjustable to at least 45 milliwatts. Frequencies shall be within the limits of 3.21. The input supply voltages shall be from 24.0 to 29.0 Vdc.

3.15.2 Relative humidity (See 4.8.2). - The unit(s) shall withstand up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. The unit(s) shall withstand the above conditions during continuous operation, intermittent operation, short term operation or exposure in a non-operating condition. Receiver quieting variations (occasional noise impulses) noted during tests shall be allowed, provided compliance with 3.17.1 is achieved.

3.15.3 Sand and dust (See 4.8.3). - The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas.

3.15.4 Vibration (mechanical) (See 4.8.4). - The unit(s) shall meet the vibration Curve III with the unit(s) in a normally mounted position by use of the Daus fasteners, no dagger pins.

3.15.4.1 Vibration (modulation) (See 4.8.4.1). - The audio output level from the receiver due to vibration modulation shall be at least 20 db below the reference level of 50 mw audio output. At major resonances, the input shall be reduced from Curve III levels to 3G with no more than one resonance allowed in each plane in which the minimum receiver quieting is at least 10 db below the reference level and one additional resonance in the fore/aft plane in which the minimum receiver quieting is at least 2 db below the reference level. The detected audio output from the transmitter due to vibration shall be at least 20 db below the reference of 90 $\pm$ 10 percent modulation except the input shall be reduced from Curve III to 3G at resonance.

3.15.5 Shock (See 4.8.5). -

3.15.5.1 Bench handling (See 4.8.5.1). - The unit(s) shall meet full specification performance following a bench handling test consisting of 4 drops from a 30 degree angle onto each face on which the unit could be placed during servicing.

3.15.5.2 Equipment (See 4.8.5.2). - The unit(s) shall meet the equipment shock test. The unit(s) shall be capable of meeting full specification performance after the 15G shock test.

3.15.6 Fungus (See 4.8.6). - The unit(s) shall show no evidence of viable fungus after being subject to 28 days of fungus test. The unit(s) shall not be required to operate.

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3.15.7 Salt fog (See 4.8.7).- The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or clogging or binding of moving parts. The unit(s) shall not be required to operate electrically.

3.15.8 Explosion (See 4.8.8). - The unit(s) shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.15.9 Acceleration (See 4.8.9). - The unit(s) shall operate within specified limits during application of 6G steady state acceleration for one (1) minute.

3.15.10 Rain, front panel (See 4.8.10). - The front panel shall withstand exposure to precipitation at variable rates to a total of 9.5 inches over a 12-hour period, with intermittent wind speeds to 35 knots, without any degradation.

3.15.11 Reliability (See 4.8.11). - The equipment shall have a reliability sufficient to provide a mean-time-between-failure of 660 hours, assuming an exponential distribution. The required reliability shall be as specified herein.

3.16 Electromagnetic Interference (See 4.9). - The emission of electromagnetic interference by the radio set and the susceptibility of the radio set to electromagnetic interference shall comply with the following requirements of MIL-STD-826 as modified herein.

3.16.1 Conducted Interference, Power lines, 30 Hz-14 kHz (Method 3001).- The narrowband power line emission limits shall be as defined in Figure 1001-1 of MIL-STD-826. Broadband power line emissions shall not exceed 140 db uA/MHz. Transient interference due to positioning the function switch from OFF to T/R and T/R to OFF are exempt from meeting these requirements.

3.16.2 Conducted Interference, Powerlines 0.014 to 100 MHz (Method 3002). - Limits shall be as defined in Figure 1. Switching transients shall be measured at 14.0 kHz with a 160 db uA/MHz limit, at 800 kHz with a 120 db uA/MHz limit, at 8 MHz with a 70 db uA/MHz limit, and at 25 MHz with 50 db uA/MHz limit. Transient broadband interference attributed to rotating the function switch from OFF to T/R and T/R to OFF are exempt from this requirement. A test method tolerance of 10 db shall be allowed for transient measurements.

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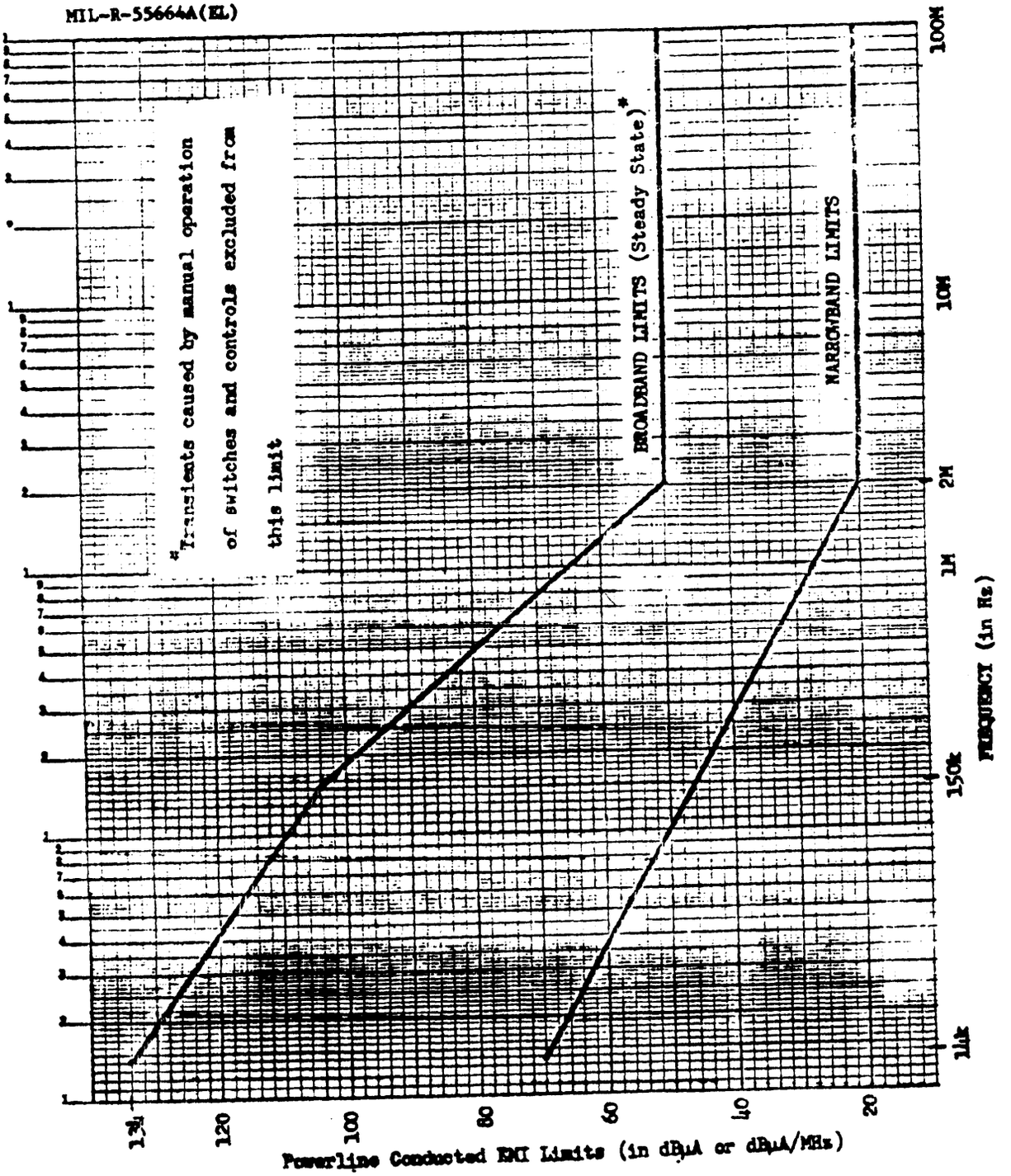


Figure 1

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3.16.3 Antenna Terminal Conducted, Low Power (Method 3003). - Narrowband emissions shall not exceed 44 db/uV and broadband levels shall not exceed 60 db uV/MHz.

3.16.4 Antenna Terminal Conducted, High RF Power (Method 3004). - Harmonic and all other spurious outputs shall be suppressed at least 66 db below the fundamental power output except that 4 spurious outputs may be suppressed less than 66 db but not less than 55 db on ten percent of the channels. A 4 db test method tolerance shall be allowed. Measurements within $f_o \pm 1\%$ shall be excluded.

3.16.5 Radiated Interference (Method 4001). -

NARROWBAND (Figures 1001-5, 1001-6 of MIL-STD-826).

a. Receive - Interference levels shall be within limits except at the first and second local oscillator frequencies and their harmonics where they shall not exceed 10 db above the limits.

b. Transmit - Interference levels shall not exceed the limits except at f_o and harmonics where they shall not exceed 35 db above the limits and at 50 MHz and its harmonics where they shall not exceed 10 db above the limit.

BROADBAND - Steady state radiated emission levels shall be within the limits shown on Figure 2.

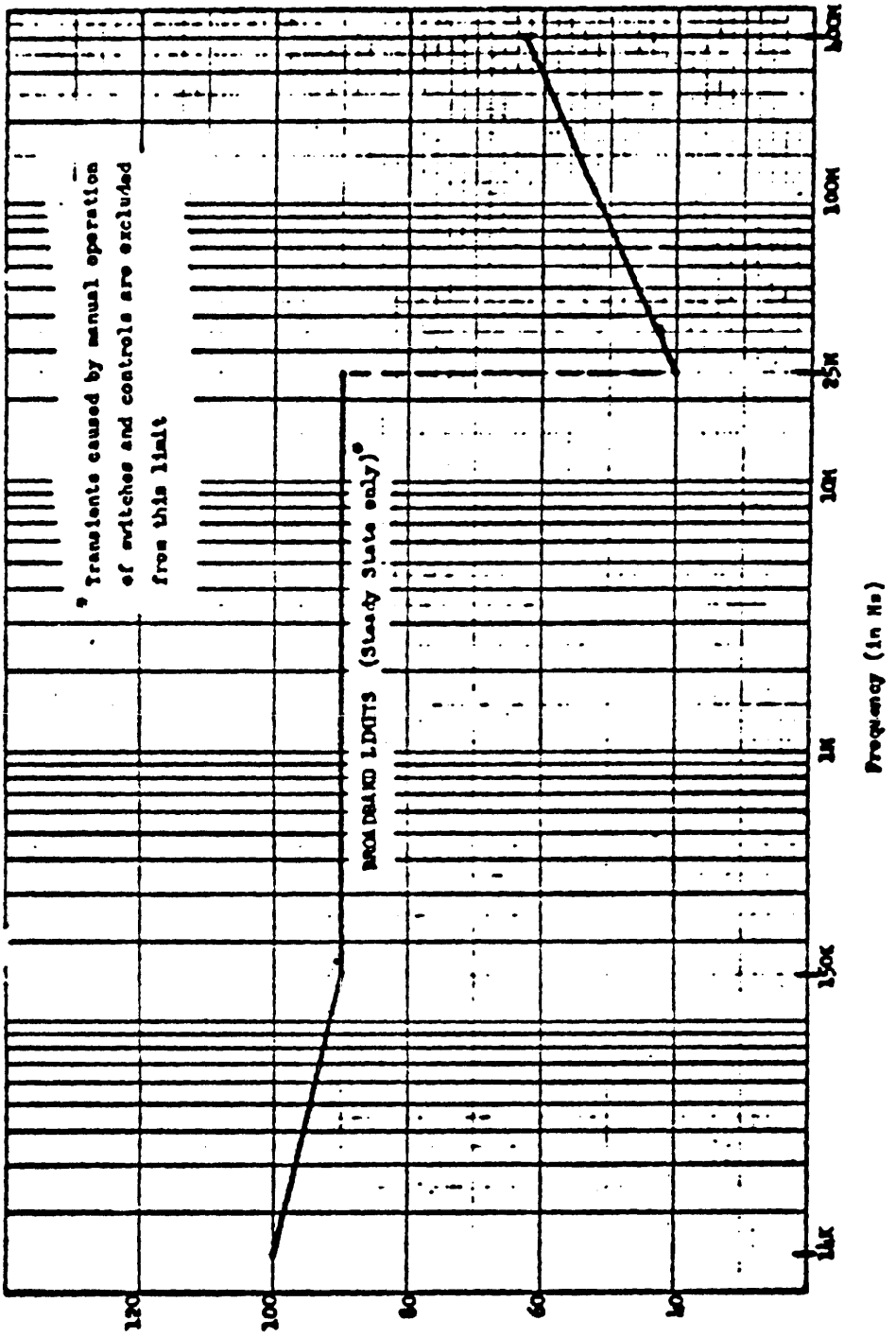
Switching transients should be measured at 14.0 kHz with a 105 db uV/m/MHz limit; at 200 kHz with a 100 db uV/m/MHz limit; at 1.2 MHz and 14.8 MHz with a 90 db uV/m/MHz limit; and at 400 MHz with a 63 db uV/m/MHz limit. Transients caused by rotating the Function switch from OFF to T/R and T/R to OFF are exempt from this requirement. A test method tolerance of 10 db shall be allowed for transient measurements.

3.16.6 Conducted Susceptibility, Powerlines 30 Hz to 150 kHz (Method 5001). - The level of injected audio signals shall be as defined in Figure 5 of MIL-STD-704. The radio set performance under these conditions shall be as follows:

a. Quieting and Internal Noise (3.17.8). - An RF Signal of 180 uV, AM modulated 30 percent at 1000 Hz, shall produce a signal-plus-noise-to-noise ratio of at least 30 db at the receiver output.

b. Squelch (3.17.2.3). - The squelch isolation loss shall be such that the audio output shall be attenuated by at least 40 db.

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Steady-State Broadband Radiated EMI Limits* (in dBμV/m/kHz)

Figure 2

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c. Squelch Operation. - The squelch control which has been preset to open on an RF signal with a signal-plus-noise-to-noise ratio of 10 db shall open at this same level with the interfering signal present.

d. Transmitter. - The transmitter noise level (See 3.18.6) shall not degrade to less than a 10 db signal-plus-noise-to-noise ratio between 30 and 150 Hz or 20 db between 150 Hz and 5 kHz or 30 db between 5 kHz and 150 kHz when interference of the specified levels is applied to the power line.

3.16.7 Conducted Susceptibility, Powerlines 150 kHz to 400 MHz (Method 5002). - The receiver performance shall be as specified for the main receiver and guard mode, except as stated below. No degradation shall be allowed on the transmitter.

a. No interference signals shall be applied within ± 50 kHz of the receiver tuned frequency.

b. When the interfering signal injected into the powerlines is in the range 6.8 - 10 MHz or at 40.1 MHz ± 50 kHz or at 59.9 MHz ± 50 kHz, specified performance must be maintained when the RF input at the antenna terminals is increased by 6 db above the sensitivity reference and the injected signal level is 250 mV.

3.16.8 Intermodulation, two-signal (Method 5003). - The level of interfering signal shall be 60 db above the sensitivity reference, except that when both signals are within ± 200 kHz of the receiver tuned frequency their level shall be 40 db above the reference level. The starting point (Δf) of ± 90 kHz shall be used for interference signal f_1.

3.16.9 Rejection of undesired signals at input terminals (Method 5005).- The requirements of 3.17.4 shall apply.

3.16.10 Spike Susceptibility (Method 5006.1). - The radio set shall comply with the requirements of MIL-STD-826. The pulse repetition rate shall be 2 pps for a period of 5 minutes or more. The spike characteristics shall be calibrated across a 5 ohm resistive (non-inductive) load.

3.16.11 Cross Modulation (Method 5007). - The level of the interfering signal shall be 60 db above sensitivity reference level.

3.16.12 Radiated Susceptibility (Method 6001). - The radio set shall not exhibit any degradation of performance, malfunction, or other undesirable effects when immersed in an RF field as specified herein, except as noted below.

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FREQUENCY RANGE	REQUIRED FIELD STRENGTH
0.014 to 2 MHz	1 Volt per meter
2.0 through 30.0 MHz	10 Volts per meter
30.0 MHz to 10 GHz	1 Volt per meter

a. Receive Mode -

(1) No RF field shall be applied within ± 50 kHz of the receiver tuned frequency.

(2) When the interfering field is within ± 50 kHz of the receiver image frequency or either IF, specified performance must be maintained when the applied RF field level is reduced by 10 db.

(3) A test method tolerance of 2 db shall be allowed in the 30 MHz to 10 GHz range for electric field receiver measurements.

b. Transmit Mode - No exceptions.

3.16.13 Radiated Susceptibility, induction field (Method 6002). - The radio set shall comply with the requirements of MIL-STD-826 except that the 400 Hz power frequency shall be used.

3.17 Electrical, Receiver (See 4.7.1 and 4.10). - Unless otherwise specified, the following shall apply to the multichannel receiver and, where applicable, to the guard/EMER receiver. The receiver shall meet all the requirements specified in 3.17.1 through 3.17.18 under standard test conditions with the equipment oriented in any position within 30 seconds after the set has been turned on. All RF input voltages specified herein are open circuit voltages from a 50 ohm source.

3.17.1 Sensitivity (See 4.10.1). - An RF signal input of 4.0 microvolts, modulated 30 percent at 1000 Hz, shall produce a signal-plus-noise-to-noise ratio of not less than 10 db when measured at a level of 50 milliwatts at the output of the main and guard/EMER receivers into the 150 ohm resistance load, except when the main receiver is tuned within ± 5.0 MHz of the guard/EMER receiver frequency, the sensitivity of the main receiver and the guard/EMER receiver shall not exceed 6.0 microvolts at 10 db signal-plus-noise-to-noise ratio. There shall be no internal quieting degradation or regeneration of sensitivity of the receiver. Under humidity, or with a combined temperature and altitude service condition including 24 and 29 volts DC, the receiver shall provide a minimum of 10 db signal-plus-noise-to-noise ratio and 50 mw with an input level of 8 microvolts.

3.17.2 Squelch (See 4.10.2). -

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3.17.2.1 Signal-to-noise operated squelch (See 4.10.2.1). - The squelch circuits in the main and guard/EMER receivers shall silence the audio output whenever there is no RF signal present at the receiver input, and it shall have the following characteristics: The squelch adjust control shall provide for the adjustment of the signal-plus-noise-to-noise required to open the squelch on signals that have a 3 to 10 db signal-plus-noise-to-noise ratio at 4.0 microvolts or less. When adjusted to the extreme counterclockwise position, it shall disable the squelch circuit for non-squelch operation. The main receiver signal-to-noise operated squelch shall be operable when the squelch switch on the front panel is operated to the "ON" position and shall be disabled when the squelch switch is operated to the "OFF" position.

3.17.2.2 Squelch overlap (See 4.10.2.2). - The ratio between the RF voltage required to unsquelch the main and guard/EMER receivers output and that voltage at which the output is again squelched shall be held to a minimum but in no case shall a change of more than 6 decibels be required.

3.17.2.3 Squelch operation (See 4.10.2.3). - The main and guard/EMER receivers shall be considered unsquelched when the audio output is within 2 db of the output obtained when the squelch is disabled. The receivers shall be considered squelched when the audio output is attenuated 50 db with no RF input signal (reference level of zero db is 50 milliwatts across 150 ohms). The main and guard/EMER receivers shall unsquelch on input signals with a signal-plus-noise-to-noise ratio of 5 through 15 db including service conditions without readjustment after the initial ambient setting of a 10 db signal-plus-noise-to-noise ratio.

3.17.2.4 Squelch pulse response (See 4.10.2.4). - The main and guard/EMER receivers shall respond (unsquelch) when receiving pulse modulated (100 percent on/off) carrier. The modulation duty cycle imposed shall be a minimum of 25 percent "ON" time. The repetition rate shall be 300 Hz (minus 20 percent) through 1000 Hz (plus 20 percent).

3.17.3 IF selectivity (See 4.10.3). - Main and Guard/EMER IF selectivity shall be as indicated:

BANDWIDTH	POINTS
56 kHz minimum	6 db
120 kHz maximum	60 db
200 kHz maximum	80 db

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(The response characteristics in the passband shall have no irregularities greater than 3.5 db from the nominal curve and shall be symmetrical about the center frequency of 9.900 MHz within 5 kHz from 6 db down to and including the -60 db points.)

3.17.4 Spurious response (See 4.10.4). - For the main receiver, there shall be no more than five (5) spurious responses beyond ± 10 MHz of the desired frequency which are less than 80 db down from the desired response, and no more than four (4) spurious responses within ± 200 kHz to ± 10 MHz which are less than 70 db down from the desired response. Image rejection shall be 80 db minimum. No spurious response outside of ± 200 kHz shall be less than 60 db down. For the guard receiver, there shall be no more than seven (7) spurious responses beyond ± 10 MHz of the desired frequency which are less than 80 db down from the desired response, and no more than five (5) spurious responses within ± 200 kHz to ± 10 MHz which are less than 70 db down from the desired response. Image rejection shall be 80 db minimum. No spurious response outside of ± 200 kHz shall be less than 60 db down, except that a maximum of three (3) shall be allowed which are no less than 55 db down beyond ± 10 MHz of the desired frequency or which are no less than 45 db down within ± 200 kHz to ± 10 MHz of the desired frequency.

3.17.5 Blocking (See 4.10.5). - When resonant signals up to one volt are present, it shall be possible to obtain a 10 db signal-plus-noise-to-noise ratio with 4.0 microvolts input not later than 0.25 second after the one volt signal has been removed. Blocking shall not occur on desired signals up to 250 millivolts.

3.17.6 Noise rejection (See 4.10.6). - With an input signal ranging from 8 to 1000 microvolts, 90 percent amplitude modulated at 1000 Hz, introduced simultaneously with pulses of the same carrier frequency, the resulting 1000 Hz audio output shall not decrease more than 1 db as a result of pulses 10 times the test voltage, nor decrease more than 2 db as a result of pulses 100 times the test voltage. The equipment shall meet the test with two different pulse signals as follows:

a.	Pulse width	10 ± 2 microseconds
	Pulse repetition rate	360 ± 60 Hz
	Rise time	1.0 microseconds
b.	Pulse width	2 ± 1 microseconds
	Pulse repetition rate	900 ± 600 Hz
	Rise time	0.1 microseconds

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(With equal unmodulated carrier and pulse amplitudes, and other test conditions as specified above, the audio output shall be at least 20 db below the reference level obtained with 90 percent amplitude modulation and normal audio gain adjustment.)

3.17.7 Desensitization (See 4.10.7). - Desensitization effects of interfering signals (modulated 30 percent at 1000 Hz) on channels 1.0 MHz or greater from, and at a level of 60 db above, the desired channel sensitivity level (10 db signal-plus-noise-to-noise ratio) shall permit a signal-plus-noise-to-noise ratio of the desired signal of at least 6 db.

3.17.8 Quieting and internal noise (See 4.10.8). - An RF signal of 180 microvolts, amplitude modulated 1000 Hz at 30 percent, shall produce a signal-plus-noise-to-noise ratio of 35 db when measured at the output of the receiver at a level of not less than 50 milliwatts. Under service conditions, the signal-plus-noise-to-noise ratio shall not be less than 29 db.

3.17.9 Automatic gain control (See 4.10.9). - Receiver audio output shall not vary more than 3 db when the RF input signal is varied over the range of 4.0 microvolts to 250,000 microvolts except between the frequencies of 225 - 255 MHz where the upper limit shall be 100,000 microvolts. The input signal shall be modulated 30 percent with a 1000 Hz tone. The receiver audio gain shall be set for a reference level of 50 milliwatts across a 150 ohm resistive load.

3.17.10 Audio frequency response (See 4.10.10). - The audio output of the receiver shall not vary more than plus 1.0 to minus 2.0 db from the level of 1000 Hz as the frequency is varied from 300 to 3000 Hz with the amplitude modulation held constant at 30 percent. The response shall be down from the 1000 Hz level by at least 6 db at 100 Hz, and 30 db at 12,000 Hz with the amplitude modulation held constant at 30 percent.

3.17.11 Audio adjust (See 4.10.11). - The main audio control shall be capable of setting the audio output level between +3 db and -20 db from a 50 milliwatt audio reference when terminated into a 150 ohm load. The auxiliary audio control shall limit the maximum output of the receiver to 50 milliwatts (-0 db, +3 db). When the main audio control is in the full counterclockwise position, an audio output level of 0.5 milliwatts (+0 db, -6 db) shall be provided at standard test conditions.

3.17.12 Audio distortion (See 4.10.12). - When the receiver is delivering a level of 50 mw to the 150 ohm resistive load, the total audio distortion shall not exceed 7 percent for either the main or the guard/EMER receivers. This shall be met for any level of modulation up to 30% and RF carrier levels up to 250 millivolts and modulation frequencies from 300 through 3000 Hz except between the frequencies of 225 - 255 MHz where the upper limit shall be 100 millivolts.

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3.17.13 Homing/DF aux circuit (See 4.10.13). - When a 1000 microvolt signal, that is modulated 30 percent at 1000 Hz, is applied at the 50 ohm antenna input circuit, the auxiliary output level shall be at least 0.25 volts. When the modulation frequency is varied from 70 to 6000 Hz, the output level for any frequency in the range shall be within plus or minus 4 db of the output obtained at 1000 Hz. The auxiliary audio output shall be in phase plus or minus 20 degrees with the positive portion of the RF signal envelope when the RF input signal is modulated 30 percent at 200 Hz. When Radio Set AN/ARC-116( ) is placed in the homing/DF mode, resonant signals that are present across the 50 ohm antenna input (which are varied in amplitude from 4.0 microvolts to 0.5 volts, and which are also keyed at rates between 100 and 400 Hz) shall provide at least 10 db change in amplitude at the keyed rate across the homing/DF output termination when the input signal at the antenna terminals is varied 13 db at the same keyed rate.

3.17.14 Receiver self test (See 4.10.14). - With the self test switch depressed, a nominal 800 Hz tone identification signal shall be present at the main receiver audio output. After depression, the test button shall not stick.

3.17.15 Maximum RF input to receiver (See 4.10.15). - Under all combination of settings of the controls, the receiver shall operate as specified after being subjected to an RF input of 10 volts open circuit peak voltage from a 50 ohm source.

3.17.16 Spurious quieting (See 4.10.16). - The RF signal input to the receiver required to produce a 10 db signal-plus-noise-to-noise ratio shall not exceed levels specified below over that required in 3.17.1 on the following channels:

GREATER THAN 40 db	LESS THAN 40 db	LESS THAN 30 db	LESS THAN 20 db	LESS THAN 10 db	LESS THAN 10 db
274.65	300.00	230.20	233.10	230.00	310.15
	366.20	260.15	236.80	235.15	310.20
		335.15	241.75	238.45	320.00
		350.00	246.40	240.00	320.05
			250.00	243.40	330.05
			260.00	244.70	333.40
			270.00	245.05	337.95
			280.20	270.05	340.00
			290.00	280.00	343.40
			295.05	286.80	345.05
			330.00	293.40	360.00
			330.20	306.80	360.15
			336.80	310.00	370.00
					390.00

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3.17.17 Synthesizer operation (See 4.10.17). - The synthesizer VCO, switch filter and preset dividers shall operate, when the receiver is in the receive mode, throughout the frequency band.

3.17.18 Audio output circuit (See 4.10.18). - Under all combination of settings of the controls, the receiver shall operate as specified after being subjected to both an open and short circuit load impedance.

3.18 Electrical, transmitter (See 4.11). - Unless otherwise specified, the transmitter, including EMER, shall meet the requirements specified in 3.18.1 through 3.18.10 under standard test conditions with the equipment oriented in any position, within 30 seconds after the radio set (in receive mode) has been turned on.

3.18.1 Audio input circuits (See 4.11.1). - The audio input impedance shall be 150 ohms plus or minus 10 percent over the audio frequency range and shall be ungrounded.

3.18.2 Power output, RF (See 4.11.2). - The unmodulated RF power output shall be not less than 8 watts nor greater than 15 watts. The unmodulated RF power output shall not be less than 4 watts with 24.0 volts dc input.

3.18.3 Modulation (See 4.11.3). - An audio input of zero dbm (one milliwatt across 150 ohms) at 1000 Hz shall produce 70 to 99 percent amplitude modulation of the carrier.

3.18.4 Frequency response, audio (See 4.11.4). - With the audio input level to the equipment held constant at the level which produces 30 percent modulation at 1000 Hz, the modulation produced by an audio input frequency from 300 to 3000 Hz shall be within plus 1.0 db to minus 2.0 db, of the modulation produced when the carrier is modulated at 1000 Hz. The modulation produced shall be down from the 1000 Hz modulation level by at least 6 db at 50 Hz, and 6 db at 12,000 Hz, with the input level constant.

3.18.5 Modulation distortion (See 4.11.5). - With any audio input frequency from 300 through 3000 Hz, and with an audio input of zero dbm (one milliwatt across 150 ohms), the distortion shall be nominally seven percent and in no case shall exceed twenty (20) percent over the temperature range of -25°F through +131°F. Under temperature-altitude and humidity service conditions, and from any input voltage between 24.0 and 29.0 volts (excluding 27.5 volts), distortion shall not exceed thirty percent.

3.18.6 Carrier noise level (See 4.11.6). - The detected audio output voltage obtained from an unmodulated carrier shall be 40 db or more below the detected output voltage obtained from a carrier modulated with 1000 Hz, at a level of zero dbm.

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3.18.7 Sidetone (See 4.11.7). - The presence of a sidetone signal shall be an indication that radio frequency power is being delivered to the load. When the RF power output falls to a level below two (2) watts, or when the VSWR exceeds 20 to 1, the sidetone shall not operate. For RF power outputs of 6 watts or more, and VSWR less than 3:1, the sidetone shall operate. The sidetone output level at 30 percent modulation shall be 6 db nominal (at least 3 db but not more than 10 db) below the receiver output. The total audio distortion of the sidetone output shall be 10 percent or less.

3.18.8 Continuous Transmit (See 4.11.8). - The transmitter shall transmit for one continuous hour under standard test conditions without power output degradation below the limit specified in 3.18.2 and without any receiver performance degradation below the limit specified in 3.17.1.

3.18.9 Retransmission (See 4.11.9). - With the unit connected to another radio set, bidirectional transfer of information shall be accomplished by receiving information on one unit and transmitting the information on another unit. Retransmission in either direction shall be accomplished. When a radio set is operated in combination with a companion radio set in the retransmission mode of operation, the amount of distortion present in the detected audio output shall not exceed 12 percent.

3.18.10 Transmitter output circuit (See 4.11.10). - Under all combination of settings of the controls, the transmitter shall operate as specified after being subjected to both an open and short circuit load impedance.

3.19 Frequency range (See 4.12). - The unit(s), shall operate on all of the 3500 distinct channels including guard/EMER. The fixed frequency guard receiver shall operate on all channels from 238 MHz through 248 MHz, when employing the proper crystals. In the EMER position, the unit shall transmit on 243.00 MHz within the frequency tolerance of 3.21.

3.20 Frequency selection (See 4.13). - The required time for the equipment to become fully operable after selection of a channel shall not exceed one (1) second.

3.21 Frequency accuracy (See 4.14). - The operating frequency of the unit(s) shall be within ± 8.0 kHz of the dial frequency within 30 seconds after turn-on with a 27.5 Vdc input applied to the unit. The operating frequency shall be within ± 16.0 kHz of the indicated frequency within 1.0 minute after turn-on with input voltage levels of 24.0 and 29.0 volts dc applied, over a temperature range from -65°F to -25°F. Under temperature-altitude and humidity service conditions, the operating frequency shall be within ± 16.0 kHz of the indicated frequency within 1.0 minute after 27.5 Vdc input is applied.

number of test samples

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sufficient

3.22 Primary DC supply (See 4.15). -

3.22.1 Input power (See 4.15.1). - With 27.5 Vdc input aircraft power as defined in MIL-STD-704, the overall power input (excluding panel lighting) to the unit(s) with 50 mw audio output in the receiver mode shall be less than 21 watts and in the transmit mode with 30 percent modulation the unit power shall be less than 85 watts.

3.22.2 Normal voltage (See 4.15.2). - The unit(s) shall meet the specification requirements under standard test conditions for any primary supply voltage between 24.0 and 29.0 Vdc, except that with 29.0 Vdc input, the RF power output shall not be less than 10 watts nor greater than 15 watts.

3.22.3 Low voltage (See 4.15.3). - With an input of 21.0 Vdc, the transmitter shall be operable and the receiver sensitivity shall be not more than 8.0 microvolts for a 10 db signal-plus-noise-to-noise ratio.

3.22.4 Emergency voltage (See 4.15.4). - The unit(s) shall not be damaged by emergency (17 Vdc) power supply conditions, per MIL-STD-704.

3.22.5 Power source transient (See 4.15.5). - The unit(s) shall not suffer permanent damage when transient voltages up to 80 volts are on the primary power line, per MIL-STD-704, Category "B".

3.22.6 Power supply protection (See 4.15.6). - The radio set power supply shall be protected from any short circuit caused by the grounding of any of the primary dc voltage lines (+5, +9, +15 and +18 vdc) and shall not cause secondary failures in the radio set. A quick means shall be provided to restore the radio set to full performance once the short circuit is removed.

3.23 EMER Transmit & Receive (See 4.16). - With the function switch in the EMER position, the unit(s) shall transmit and receive on the guard/EMER frequency and meet the guard receiver and emergency transmitter requirements herein. The multi-channel receiver shall be disabled.

3.24 X-Mode (See 4.17). -

3.24.1 Input, wideband (See 4.17.1). - With a 20 kHz square wave signal at 12 volts peak-to-peak clamped to ground applied to transmitter audio input through either a 100 ohm or 780 ohm source impedance, the modulation level at the transmitter output shall be between 70 to 99 percent under standard test conditions and under service conditions. The transmitter shall process a plain cipher control signal of "ground" in cipher and a + 28 Vdc in plain. The wideband input impedance shall be greater than 100 ohms.

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3.24.2 Output, wideband (See 4.17.2). - The wideband output from the receiver shall be a voltage of at least 0.5 volts peak-to-peak across 50,000 ohms in parallel with a 1000 picofarad capacitor.

3.24.3 Output, narrowband (See 4.17.3). - With audio control set at maximum clockwise position, the narrowband output from the receiver shall deliver a minimum of 2.10 Vrms across 600 ohms in parallel with the 150 ohm main audio load.

3.24.4 Output, guard channel (See 4.17.4). - While receiving a 1 kHz reference tone level modulated 30 percent the guard channel only output from the receiver shall deliver a minimum of 2.73 Vrms across 600 ohms. When not receiving digital information, guard channel shall be received through the narrowband output (guard plus main receiver).

3.24.5 Output, receiver (See 4.17.5). - The unit(s) shall operate in narrowband or plain mode output, wideband or cipher output and guard channel only output modes and shall receive plain traffic while in the cipher mode. While receiving a cipher signal the filtered wideband information shall be a minimum of 40 db below the level of the guard channel only output operation. Application or removal of a 15 kHz square wave signal on the main channel shall result in less than 3 db degradation to the received 1 kHz tone on the guard channel only output.

3.24.6 Operational, X-Mode (See 4.17.6). - The unit(s) shall operate and interface with the TSEC/KY-28.

3.25 Turn Around Time (See 4.24). -

3.25.1 Turn Around Time, Transmit to receive (See 4.24.1). - With the transmitter producing full rated power output, the receiver shall produce a steady state full performance output with normal squelch operation within 600 milliseconds after the transmitter key line is deactivated.

3.25.2 Turn Around Time, Receive to transmit (See 4.24.2). - When the transmitter key line is activated, the transmitter power output shall rise to the full rated power output with full modulation within 400 milliseconds.

3.26 Burn-In/debugging (See 4.6). - Each unit on order shall meet the requirements herein, without subsequent processing, after subjection to the Burn-In/debugging as specified.

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3.27 Interchangeability (See 4.19). - Like units, assemblies, sub-assemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable. Items which are covered by specified electrical performance requirements shall meet such specified performance requirements prior to installation into the unit(s). After installation of these items and prior to submission of unit(s) for the quality performance specifications, adjustments shall be made, if required, to permit the unit(s) to meet full equipment specification requirements. Alteration of any items, or the unit(s) shall not be permitted.

3.28 Operational inspection (See 4.20). - Immediately prior to packaging and after applicable processes, inspections and tests are completed, each unit(s) shall be subjected to the operational inspection as specified after being set for the following normal conditions:

a. The guard receiver frequency shall be set for 243.00 MHz.

b. The internal audio output adjustment control shall be adjusted for 50 mw audio output with the external audio control set fully clockwise and with a 4.0 microvolt RF input signal modulated 90 percent at 1000 Hz.

c. The internal main and guard squelch controls shall be set to open at 10 db signal-plus-noise-to-noise ratio at frequencies of 312.05 and 243.00 MHz respectively with a maximum of 4.0 microvolts RF input signal modulated 30 percent at 1000 Hz.

d. The internal guard audio output adjustment control shall be adjusted for 2.73 Vrms across 600 ohms with a 4.0 microvolts RF input signal modulated 90 percent at 1000 Hz.

3.29 Technical literature and spare parts (See 6.2). - Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the equipment furnished on order.

3.30 Workmanship (See 4.18). - The equipment shall be manufactured and assembled in accordance with the equipment drawings (See 3.2) and the following requirements and paragraphs herein:

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MIL-STD-454:

Soldering	Requirement 5
Workmanship	Requirement 9

Paragraphs herein:

Cleaning	3.4
Controls	3.6
Securing of parts	3.8
Soldering	3.10
Wiring, cabling, bonds and grounds	3.11
Tropicalization of materials	3.14

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality Conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.22 and 4.23)

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4.3 First Article inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, B and C (See Tables I, II and III respectively). The First Article inspection will normally be performed in this order: Sample 1: (1) Temperature/altitude; (2) Humidity; (3) Vibration; (4) Rain, front panel; and (5) Sand and dust; Sample 2: (1) Vibration; (2) Temperature-altitude combination; (3) Humidity; (4) Shock, bench handling and equipment; and (5) Salt fog; Sample 3: (1) Electromagnetic interference; and (2) Explosion proof; Sample 4: (1) Acceleration; (2) Fungus; Samples 5 through 13: Reliability. Other First Article inspections may precede, follow, or be interspersed among the foregoing. (See 3.26)

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents, as part of the inspection before preparation for delivery:

ITEM	WHERE REQUIRED
Finish protective	3.7
Marking	3.9
Conformal coating	3.12
Semiconductors (transistors & diodes)	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to Burn-In/debugging (See 3.26) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-In/debugging shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (See 6.8). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot shall be subject to verification utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQLs indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection in 4.20 shall be last. (See 6.4)

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TABLE I - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.30	4.18		
Modules (Fluck-out):				
Switched Filters Circuit Card			0.65%	2.5%
Divider, 10 MHz oscillator and detector circuit card			0.65%	2.5%
Voltage Controlled Oscillator Circuft Card			0.65%	2.5%
59.9 MHz IF Circuit Card-Guard			0.65%	2.5%
59.9 MHz IF Circuit Card-Main			0.65%	2.5%
Modulator/Carrier Control Circuit Cards			0.65%	2.5%
IF Amplifier Circuit Card			0.65%	2.5%
Second IF, Detector & AGC Amplifier Circuit Card			0.65%	2.5%
Squelch Circuit Card			0.65%	2.5%
Audio Amplifier Circuit Card			0.65%	2.5%
Assemblies:				
Interconnect Board and Harness Assembly			0.65%	2.5%
Tuner Matrix Deck Circuit Card Assys			0.65%	2.5%
Indicating Light, Transmitting Panel Assembly			0.65%	2.5%
Frequency Selector Assy			0.65%	2.5%
Power Supply Circuit Card Assy			0.65%	2.5%
Modulator/Carrier Control Assy			0.65%	2.5%
RF Front End and Electronic Component Assy			0.65%	2.5%
Electrical Equipment Chassis, Wired			0.65%	2.5%
Exciter Assy			0.65%	2.5%
Driver-Power Amplifier Assy			0.65%	2.5%
Front Section (less assemblies)			0.65%	2.5%
Rear Section (less assemblies)			1.0%	4.0%
Front Section (assembled)			1.0%	4.0%
Rear Section (assembled)			1.0%	4.0%
Radio Set AN/ARC-116( ) (assembled)				

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TABLE I - Group A Inspection (Cont'd)

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical (Cont'd)	3.30	4.18		
Electrical				
Receiver:				
Sensitivity	3.17.1	4.10.1		
Signal to noise oper. squelch	3.17.2.1	4.10.2.1		
Squelch overlap	3.17.2.2	4.10.2.2		
Squelch operation	3.17.2.3	4.10.2.3	*1.0% for	
Squelch pulse response	3.17.2.4	4.10.2.4	combined group.	
Quieting & internal noise	3.17.8	4.10.8		
Audio distortion	3.17.12	4.10.12		
Receiver self test	3.17.14	4.10.14		
Synthesizer operation	3.17.17	4.10.17		
Transmitter:				
Power output, RF	3.18.2	4.11.2		
Modulation	3.18.3	4.11.3		
Modulation distortion (amb.only)	3.18.5	4.11.5		
Frequency selection	3.20	4.13		
Retransmission	3.18.9	4.11.9		
Frequency accuracy	3.21	4.14		
Operational inspection	3.28	4.20	0.1%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.3 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

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TABLE II - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Receiver:			
IF selectivity	3.17.3	4.10.3	6.5% for the combined group. (Rec & Trans)
Automatic gain control	3.17.9	4.10.9	
Audio frequency response	3.17.10	4.10.10	
Audio adjust	3.17.11	4.10.11	
Homing/DF aux. circuit	3.17.13	4.10.13	
Maximum RF input to receiver	3.17.15	4.10.15	
Output, wideband	3.24.2	4.17.2	
Output, narrowband	3.24.3	4.17.3	
Output, guard channel	3.24.4	4.17.4	
Output, receiver	3.24.5	4.17.5	
Transmitter:			
Audio input circuits	3.18.1	4.11.1	
Frequency response, audio	3.18.4	4.11.4	
Carrier noise level	3.18.6	4.11.6	
Sidetone	3.18.7	4.11.7	
Input, wideband	3.24.1	4.17.1	
Turn Around Time	3.25	4.24	6.5%
Rain, front panel	3.15.10	4.8.10	6.5%
EMER Transmit & Receive	3.23	4.16	6.5%
Interchangeability (Mechanical)	3.27	4.19	6.5%
Interchangeability (Electrical)	3.27	4.19	6.5%

4.5.4 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

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TABLE III - Group C Inspection

Inspection	Reqt Para	Insp Para
Subgroup I:		
Spurious response	3.17.4	4.10.4
Blocking	3.17.5	4.10.5
Noise rejection	3.17.6	4.10.6
Desensitization	3.17.7	4.10.7
Audio output circuit	3.17.18	4.10.18
Continuous transmit	3.18.8	4.11.8
Transmitter output circuit	3.18.10	4.11.10
Spurious quieting	3.17.16	4.10.16
Frequency range	3.19	4.12
Input power	3.22.1	4.15.1
Normal voltage	3.22.2	4.15.2
Low voltage	3.22.3	4.15.3
Emergency voltage	3.22.4	4.15.4
Power source transient	3.22.5	4.15.5
Power supply protection	3.22.6	4.15.6
Subgroup II:		
Temperature-altitude combination	3.15.1	4.8.1
Relative humidity	3.15.2	4.8.2
Vibration (mechanical)	3.15.4	4.8.4
Vibration (modulation)	3.15.4.1	4.8.4.1
Subgroup III:		
Sand and dust	3.15.3	4.8.3
Bench handling	3.15.5.1	4.8.5.1
Equipment	3.15.5.2	4.8.5.2
Fungus	3.15.6	4.8.6
Salt fog	3.15.7	4.8.7
Explosion	3.15.8	4.8.8
Acceleration	3.15.9	4.8.9
Operational X-Mode	3.24.6	4.17.6
Weight	3.5	4.21
Subgroup IV:		
Reliability	3.15.11	4.8.11
Subgroup V:		
Electromagnetic Interference	3.16	4.9

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4.5.4.1 Sampling for inspection. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1.1 through 4.5.4.1.5. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s). Selection of samples shall be as specified and within 45 calendar days (within 60 calendar days for Subgroup IV) after selection the samples shall have been subjected to and completed the Group C inspections. (See 6.3)

4.5.4.1.1 Subgroup I. - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent inspection, two units per month shall be selected.

4.5.4.1.2 Subgroup II. - For this subgroup, two units (these units may be the units selected for above Subgroup I), from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every two months shall be selected.

4.5.4.1.3 Subgroup III. - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected every four (4) months, except that fungus, acceleration, explosion, sand and dust, and operational X-mode shall be selected when one-half of the units on order have been accepted by the Government QA representative.

4.5.4.1.4 Subgroup IV. - For this subgroup, nine (9) units shall be selected from the first twenty (20) production units. For subsequent Group C inspections, nine (9) units shall be selected from every two (2) months production; the samples shall be selected from different lots so that the units will be representative of the two (2) months production.

4.5.4.1.5 Subgroup V. - For this subgroup, one (1) unit from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspections, one (1) unit shall be selected every four (4) months. These units shall not have been subjected to 4.5.4.1.2 through 4.5.4.1.4.

4.5.5 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units

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shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B will be withheld until the Group C reinspection results have shown that the corrective action was effective. For other non-compliance, see 4.8.11.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-In/debugging. - Each unit on order shall be subject to a Burn-In period at test level E, 4.1.6 of MIL-STD-781, except that the duration of the vibration shall be at least 50 minutes of each on-time hour and input voltage cycling is not required. The duty cycle shall be receive five (5) minutes and transmit one (1) minute except when taking measurements. The Burn-In period shall consist of twelve temperature cycles. Each temperature cycle shall be in accordance with the "Alternative Method of Temperature Cycling", 5.2.3.2 of MIL-STD-781, except that prior to measurements the equipment shall be "OFF" during the cooling period and that the time period of 2 hours after stabilization shall not apply. After equipment stabilization at the lower extreme temperature, the equipment shall be turned on and a complete set of measurements taken and recorded. During temperature excursion, a complete set of measurements shall be taken and recorded. After equipment temperature stabilization is achieved at the upper extreme temperature, a complete set of measurements shall again be taken and recorded. The electrical parameters to be measured are those used for the Reliability Test (4.8.11.7). Measurements shall be taken during periods of vibration and no-vibration. The test records shall indicate the environment at each reading (e.g. temperature + vibration). Measurements taken at temperature excursion need only be made at one tuned frequency. All measurements shall be taken during the first, one of the middle, and last temperature cycles, and shall be at different tuned frequencies during each cycle so that all groups (4.7.2) are covered during the Burn-In period. If measurements taken during a period of no-vibration are not completed within ten minutes, the vibration shall be switched off until the measurements are completed. Each unit shall also be monitored for 30 seconds duration at each measurement to determine if intermittencies exist. An intermittency will be considered to be a failure. Each unit which fails during the Burn-In period must be repaired and returned to test until it successfully survives the entire Burn-In period without failure. Failures which occur during the Burn-In shall be noted and reported, but shall not be counted towards the establishment of an MTBF. Prior to use on contract, the test chambers shall be set-up and loaded in the same manner as to be used during production and

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(a) thermal survey(s) conducted to verify that the "Test Chambers" and "Thermal Survey" requirements of MIL-STD-781 and this Burn-In can be met. The thermal survey(s) shall include the time-temperature relationship between each and all equipments in the chambers(s) and the chamber(s) air. The results of the thermal survey(s) are to be included in the Burn-In test procedures. The temperature cycle shall not be less than four (4) hours nor more than eight (8) hours. Except as otherwise noted, test facilities shall be in accordance with 5.3 of MIL-STD-781. Burn-In fixtures, test procedures and temperature cycle shall be approved by the procuring activity prior to use. The axis of vibration shall be from the top to the bottom of the unit(s). This test shall be performed to determine compliance with 3.26.

4.7 Standard test conditions. - Unless otherwise specified, inspections shall be performed at room temperature of 77° ± 10°F and humidity not to exceed 90 percent under the following conditions:

a. Input voltage: 27.5 (+0.5,-0.0) volts dc.

b. Duty cycle: One (1) minute transmit, five (5) minutes receive.

c. Warm-up time: Not more than 30 seconds.

d. Function switches: T/R, unless testing guard, (Main receiver tuned more than 5 MHz from the guard frequency) D/F, RETRAN, or EMER. Squelches: OFF.

e. Load impedance: RF 50 ± 5% ohms resistive; audio 150 ± 10% ohms resistive ungrounded.

f. Audio output level: 50 mw (with 30 percent 1000 Hz modulation applied).

g. Input Impedances: RF 50 ohm: audio 150 ohms ungrounded.

h. Audio input level: 0 dbm at 1000 Hz into 150 ohms ungrounded.

4.7.1 Standard test signal. - The standard test signal for the receiver is 4.0 microvolts open circuit from a 50 ohm source with 30 percent modulation at 1000 Hz. Generators calibrated in closed circuit voltages may employ a 6 db pad, or if their source impedance is already 50 ohms, the output voltage reading is corrected to open circuit values by multiplying by two (2).

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4.7.2 Standard test frequencies. - These frequencies shall be utilized as specified, and random frequency selection shall be made for each unit from each frequency group during inspections: Group 1 (225.00 - 268.40 MHz), Group 2 (268.45 - 311.80 MHz), Group 3 (311.85 - 355.15 MHz), Group 4 (355.20 - 399.95 MHz).

4.8 Service conditions. - Unless otherwise specified, the service conditions tests shall be conducted in accordance with MIL-T-5422 for Class 1 equipment and as defined in 3.15. Electrical tests and inspections shall be conducted as specified in Table IV. No defects shall be allowed. The duty cycle shall be five minutes receive and one minute transmit.

4.8.1 Temperature-altitude combination. - Operate the unit(s) under standard test conditions and then subject the unit(s) to the operating, non-operating, altitude and temperature range with supply voltages from 24 to 29 Vdc as specified in MIL-T-5422. In addition, at temperatures of -25°F and +131°F and at atmospheric pressure, the required electrical test shall be run at 27.5 Vdc. These tests shall be made to determine compliance with 3.15.1.

4.8.2 Relative humidity. - The unit(s) shall be subjected to the humidity test of MIL-T-5422 to determine compliance with 3.15.2. The unit(s) shall meet the specified electrical operating tests at the start of the fourth and the start of each subsequent cycle. At the end of this test, the warm-up time for the equipment shall be 30 seconds and the unit(s) shall meet the specified electrical tests.

4.8.3 Sand and dust. - The unit(s) shall be subjected to the sand and dust test of MIL-STD-810, Method 510/1 for compliance with 3.15.3.

4.8.4 Vibration (mechanical). - The unit(s) shall be subjected to the vibration test of MIL-T-5422 to determine compliance with 3.15.4.

4.8.4.1 Vibration (modulation). - At resonances, with inputs as specified but not greater than 3G, measure the receiver audio output and the transmitter detected audio output for compliance with 3.15.4.1. The vibration time for this test shall be held to a minimum, and shall not be counted toward the total time requirement of 4.8.4.

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TABLE IV

TESTS	FREQ. MHz	VIB.		TEMP. ALT.		SHOCK, EQUIP-MENT		ACCEL.		HUMID		SAND & DUST		EXPLO-SION	
		Before & After	During	Before & After	During	Before & After	During	Before & After	During	Before & After	During	Before & After	During	Before & After	During
Sensitivity 4.10.1	227.05 264.65 302.40 395.75 Guard EMER	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Squelch operation 4.10.2.3		X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Quieting and internal noise 4.10.8		X	N/A	X	N/A	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Receiver self test 4.10.14		X	N/A	X	X	X	N/A	X	N/A	X	N/A	X	N/A	X	N/A
Input, wideband 4.17.1		X	N/A	X		X	N/A	X	N/A	X	X	X	N/A	X	N/A
Power output RF 4.11.2	227.05 264.65 302.40 395.75 Guard EMER	X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Modulation distortion 4.11.5		X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Frequency accuracy 4.14		X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Output, wideband 4.17.2		X	X	X	X	X	N/A	X	N/A	X	X	X	N/A	X	N/A
Duty cycle of 4.8	5:1		X		X							X			
Vibration(modulation) 4.8.4.1	To be conducted during Vibration only.														
Operational 4.20 (on one frequency)	To be conducted during Acceleration and Explosion.														
Mechanical and visual inspection and also operation of controls 4.18	To be conducted before and after each Service conditions test.														

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4.8.5 Shock. -

4.8.5.1 Bench handling. - The unit(s) shall be assembled for servicing and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing, to determine compliance with 3.15.5.1:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using the other edges of the same horizontal face as axes for a total of four drops for each axis.

b. Repeat a. above, with the assembly resting on the other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests of 4.10.1, 4.11.2 and 4.14 shall be performed before and after shock at a supply voltage of 27.5 volts.

4.8.5.2 Equipment. - The unit(s) shall be subjected to the shock tests of MIL-T-5422. These tests shall include the 30 G Crash Safety Test. After the 15 G tests, the unit(s) shall be tested as required in 4.8 to determine compliance with 3.15.5.2.

4.8.6 Fungus. - The unit(s) shall be subjected to 28 days of fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified to determine compliance with 3.15.6. The unit(s) shall not be required to operate.

4.8.7 Salt fog. - The unit(s) shall be exposed in its normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. The unit(s) shall not be opened. The unit(s) shall not be required to operate electrically. All external moveable controls, access covers, and the test button shall be operated mechanically. These tests shall be performed to determine compliance with 3.15.7.

4.8.8 Explosion. - The unit(s) shall be tested for explosion in accordance with Procedure I of MIL-T-5422 to determine compliance with 3.15.8.

4.8.9 Acceleration. - The unit(s) shall be tested in accordance with Method 513, Procedure II of MIL-STD-810. The unit(s) shall be subjected, in turn, in each direction along each of its three orthogonal axes. This test shall be performed to determine compliance with 3.15.9.

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4.8.10 Rain, front panel. - With the unit(s) completely assembled and in a sealed chassis case (test fixture), the front panel of the unit(s) fully exposed shall be subjected to the rain test of Method 506, Procedure 1 of MIL-STD-810. During the 30 minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period. After completion of the rain test, the unit(s) shall be removed from the sealed chassis (test fixture) subjected to, and meet, the tests of 4.10.1, 4.11.2. and 4.14. Within 10 minutes after completing the electrical test, and no more than thirty minutes after completing the rain test, open the unit(s); there shall be no evidence of leakage. This test shall be performed to determine compliance with 3.15.10.

4.8.11 Reliability. - The procedures and tests required by 4.8.11.1 through 4.8.11.9 shall be performed to determine compliance with 3.15.11.

4.8.11.1 Accept/reject criteria. - The accept/reject criteria for the unit(s) shall be in accordance with Test Plan III, 4.2.8.3 of MIL-STD-781, using a specified MTBF of 660 hours.

4.8.11.2 Reliability test conditions. - Reliability Testing shall commence on samples of units taken from lots which have successfully completed the Burn-In/debugging and have met Group A and B inspection. Test Level E, 4.1.6 of MIL-STD-781, shall be used for testing (except that the rate of temperature change of the thermal medium shall not be less than 1°C/minute). The duty cycle shall be receive five (5) minutes and transmit one (1) minute except when taking measurements. The total test time per unit shall be in accordance with 4.2.2.4 of MIL-STD-781 and the temperature cycle shall not be greater than 8 hours. The axis of vibration shall be from the top to the bottom of the unit(s). Test facilities shall be in accordance with 5.3 of MIL-STD-781 except as otherwise noted.

4.8.11.3 Thermal Survey. - At the beginning of the program, a thermal survey in accordance with MIL-STD-781 shall be performed on nine (9) units to determine unit temperature versus chamber air temperature.

4.8.11.4 Data recording interval. - Data shall be recorded on a daily basis, i.e., seven days per week. The first data point shall be taken during the heating cycle, starting when the unit(s) temperature reaches -25°F and readings shall continue until a complete set of parameters is taken and recorded. The remainder of the required daily measurements shall be taken starting when the unit temperature passes through +68°F and shall continue until all data are recorded. At the conclusion of each day's measurements, the units shall be set for the frequency to be read at -25°F, in order that the data may be taken the next day without the necessity of opening the chamber to set the frequency. The data taken at -25°F shall be at a different frequency every day.

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4.8.11.5 Data recording during vibration. - Measurement of each parameter on each unit shall be taken at least once during the course of the test while the unit is under 2.2G vibration. Since the minimum test time for an accept decision is 11 days, each parameter must be measured under vibration conditions within 11 days of test start. For simplicity and ease of record keeping, this may be accomplished by measuring all required parameters on one unit per day under vibration conditions. In the event the data measurements have not been completed on this unit when the vibration cycle is complete (10 minutes of every ON hour), the vibration shall be manually switched back on until the measurements have been completed. A note shall be added to the data sheet, noting that these measurements were made under vibration conditions.

4.8.11.6 Required daily measurements. -

4.8.11.6.1 Electrical parameter measurements (See 4.8.11.7). - Electrical parameters to be monitored for each unit during the reliability tests shall be as follows:

PARAMETER	NO. OF MEASUREMENTS/DAY	TEST FREQUENCY
Main Receiver Sensitivity	4	Main test frequencies per Table V.
Guard Receiver Sensitivity	1	The Guard/EMER per Table V.
Sidetone presence & Power output	5	Same frequencies used for Main Sensitivity and the Guard/EMER frequency.
Modulation	5	Same as for Power Output & Sidetone Presence.
Squelch Operation	5	The Main & Guard/EMER frequencies per Table V.
Frequency Accuracy	5	The Main & Guard/EMER frequencies per Table V.

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The frequencies and sequence which shall be utilized are tabulated in Table V.

TABLE V

WEEK OF TEST	TEST FREQUENCIES (MHz)				
	MAIN				GUARD/EMER*
1	226.00	269.25	312.05	359.50	243.00
2	237.50	273.05	322.50	368.00	243.00
3	248.00	284.50	331.05	387.50	243.00
4	259.50	295.00	340.50	396.55	243.00
5	Same as Week 1				"
6	Same as Week 2				"
7	Same as Week 3				"
8	Same as Week 4				"
9 etc.	Repeat Sequence starting with Week 1.				

*When checking the guard receiver or the EMER channel, the main receiver will be set at 395 MHz.

4.8.11.6.2 Operating time measurement. - An elapsed time meter shall be used to monitor the number of operating hours accumulated for each equipment. The elapsed time meter shall be connected such that only equipment "ON" time will be recorded.

4.8.11.7 Electrical parameter requirements (See 4.8.11.6). - Electrical parameter requirements shall be those specified in 3.17.1, 3.17.2.3, 3.18.2, 3.18.3, 3.18.7, and 3.21. Measurements shall be made in the most efficient order to minimize test time. Each parameter shall be monitored continuously for at least 30 seconds for intermittencies.

4.8.11.8 Failure determination. -

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4.8.11.8.1 Failure definition. - A failure is defined as the inability of the equipment to perform its required function (including panel lamp operation). For the purpose of this test, any variation of the daily measured parameters outside the limits established in 4.8.11.7 including intermittencies shall constitute a failure.

4.8.11.8.2 Failure classification. - Each failure occurring during this test shall be classified as relevant or non-relevant. Only relevant failures shall be counted in formulating an accept-reject decision.

4.8.11.8.2.1 Non-relevant failure. - A failure shall be classified as non-relevant when analysis reveals the failure to be due to (a) improper test equipment adjustment, (b) failure of another part (dependent failure), (c) test installation error and (d) failure of test equipment of facility (e.g., Prime Power).

4.8.11.8.2.2 Relevant failure. - All failures not specified in 4.8.11.8.2.1 shall be relevant failures.

4.8.11.8.3 Failed unit(s) - On the occasion of a failure during the test, entries shall be made in the appropriate data logs, the failed unit(s) shall be removed from test and the following actions shall be performed: (a) the failure shall be confirmed, (b) the failure shall be isolated and the repair made, (c) failure analysis shall be performed and recorded, (d) repair effectiveness shall be verified. The test on the remaining units shall continue while the failed unit(s) is being repaired.

4.8.11.8.3.1 Failure confirmation. - Everything external to the failed unit(s) shall be closely examined in a timely manner to determine if a non-relevant failure exists.

4.8.11.8.3.2 Failure isolation and repair. - Failures shall be isolated to the smallest part. Repair of the failed unit(s) shall be limited to replacement of the defective assembly, subassembly, card or part, with an identical Burned-In item. The failed item(s) shall then undergo analysis to determine the cause of failure.

4.8.11.8.3.3 Failure analysis and corrective action. - All failures shall be reported and analyzed. Corrective action or recommended design changes shall be submitted to the procuring activity for approval. When pattern failures per MIL-STD-781 are observed during any reliability test of either an accept or reject decision, the contractor shall notify the procuring activity within ten (10) days of the recommended corrective action to be taken.

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4.8.11.8.3.4 Verifying effectiveness of repair. - Upon return of a unit from the repair area to the test area, following repair or corrective action, the unit shall be re-installed in the test chamber, allowed to operate one day under the Reliability Test conditions, and the required parameter measurements taken. The repair shall be considered effective if the measured parameters are within the limits specified in 4.8.11.7. In the event performance is not satisfactory, the unit shall be removed and returned to the repair area for further troubleshooting and repair. Failures and operating time accumulated during this period shall be recorded and reported but not used in determining compliance with the MTBF requirement.

4.8.11.9 Reliability demonstration compliance/non-compliance. -

4.8.11.9.1 Compliance. - When the total unit hours of test, together with the total count of relevant equipment failures result in an accept decision in accordance with the accept-reject criteria, compliance shall have been demonstrated and testing shall be terminated.

4.8.11.9.2 Non-compliance. - When the total unit hours of test, together with the total count of relevant equipment failures result in a reject decision, non-compliance shall be indicated and further shipment of units shall be discontinued.

4.8.11.9.2.1 Action resulting from non-compliance. - Immediately upon reaching a reject decision, the procuring activity shall be notified of the occurrence and the Reliability Test shall be terminated until corrective action can be implemented on the lot(s) from which the rejected test samples were drawn and all subsequent lot(s). Upon implementation of the approved corrective action, new test samples from the same lot(s) from which the test samples were selected shall be submitted for retest. The sample size submitted for this re-test may be the same (nine units) or greater than the rejected sample size. Resumption of shipment shall be withheld, pending the results of the retest. When the retest results in an accept decision and approved corrective actions have been taken, shipment of units shall be resumed, upon approval of the procuring activity. When the retest results in a reject decision, shipment of units shall be withheld and action shall be taken as described for rejected units.

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4.9 Electromagnetic Interference. - The unit(s) shall be tested to determine compliance with the requirements of 3.16 through 3.16.13. Tests shall be performed under standard test conditions as specified in 4.7 except that test frequencies for all EMI/EMC tests shall be 227.05 MHz, 263.20 MHz, 312.40 MHz and 395.75 MHz for the main receiver and transmitter. The test frequency for the guard receiver and emergency transmit operation shall be 243.00 MHz. The panel lamp power line shall be included in power line conducted interference and power line conducted susceptibility tests. When performing tests on the main receiver, the guard frequency shall be excluded as a test frequency. Tests performed on the guard receiver shall exclude the main receiver front panel frequency as a test frequency. When testing the guard receiver, the main receiver shall be set at 395 MHz. All tests procedures shall be in accordance with the methods of MIL-STD-826 except as modified herein.

a. Transient Interference Measurements, Conducted and radiated. - Transient interference measurements shall be performed in accordance with MIL-STD-826, Method 2003, Paragraph 5. This method allows measurement of the transient within two levels that are 10 db apart.

b. Rejection of Undesired Signals at Input Terminals. - This test shall be performed using a one signal generator method as follows:

(1) Test apparatus shall consist of signal generators capable of delivering the required levels over the applicable frequency range, filters to eliminate signal generator spurious outputs, and equipment to monitor test sample performance.

(2) Set up the test instrumentation as shown on Figure 5005-1 of MIL-STD-826 except that only one signal generator shall be used. Impedance match shall be maintained between signal generator and filter and between filter and test sample.

(3) Adjust the output of the signal generator, tuned to F_O, to give the standard reference output on the test sample. The signal generator shall be modulated 30 percent by 1000 Hz sine wave throughout the test procedure.

(4) Increase the output of the signal generator to the specified limit above that obtained in (3) above, and scan the signal generator over the required frequency range. Check all responses to assure that a spurious response is being measured and not an intermodulation product. Care shall be taken not to mistake harmonics and spurious of the signal generator falling on F_O for spurious responses of the test sample. It is possible to have spurious responses at harmonics of F_O which are not due to generator harmonics.

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(5) When a true spurious response is obtained, decrease the generator output until the standard response is obtained. Record the frequency and level required to obtain the standard reference output. Calculate the difference in db between this level and that obtained in (3). This is the spurious response rejection.

c. Radiated Interference. - Emanation levels above 200 MHz shall be measured, using the Spiral Log Conical Antenna factors given in Figures 3 and 4.

d. Radiated Susceptibility, Electric Field. - In the frequency range above 200 MHz the radiated field shall be calibrated using the Spiral Log Conical Antenna factors given in Figures 3 and 4.

e. Radiated Susceptibility, Induction Field. - Test shall be in accordance with MIL-STD-826, Method 6002, except that 400 Hz power frequency shall be used for the susceptibility current.

4.10 Electrical, receivers. - All measurements are to be performed under standard test conditions unless otherwise noted and procedures are the same for multichannel and guard/EMER receivers.

4.10.1 Sensitivity. - Apply a standard test signal (4.7.1) to the receiver input. Record the audio output ratio (decibels) between modulated signal to unmodulated signal. Perform this test at twenty (20) frequencies, five (5) selected from each of the frequency groups (4.7.2) for the multichannel receiver. Repeat this test at 243.00 MHz for both the guard and EMER functions. These tests shall be performed to determine compliance with 3.17.1.

4.10.2 Squelch. -

4.10.2.1 Signal to noise operated squelch. - Under the standard test conditions, but without input signal, operate the receiver and verify that the squelch is disabled when the squelch control switch is OFF. The squelch will also be disabled when the switch is ON and the internal control is in full counterclockwise position. Apply a 312.05 MHz input signal amplitude modulated 30 percent at 1000 Hz. Set the input signal level for a signal-plus-noise-to-noise ratio of 3 db and then 10 db measured at the receiver output. Adjust the internal squelch control at each of these sensitivity levels and verify that the squelch can be enabled and disabled by the input signal. Repeat the above test for the guard/EMER receiver at the guard frequency. These tests shall be performed to determine compliance with 3.17.2.1.

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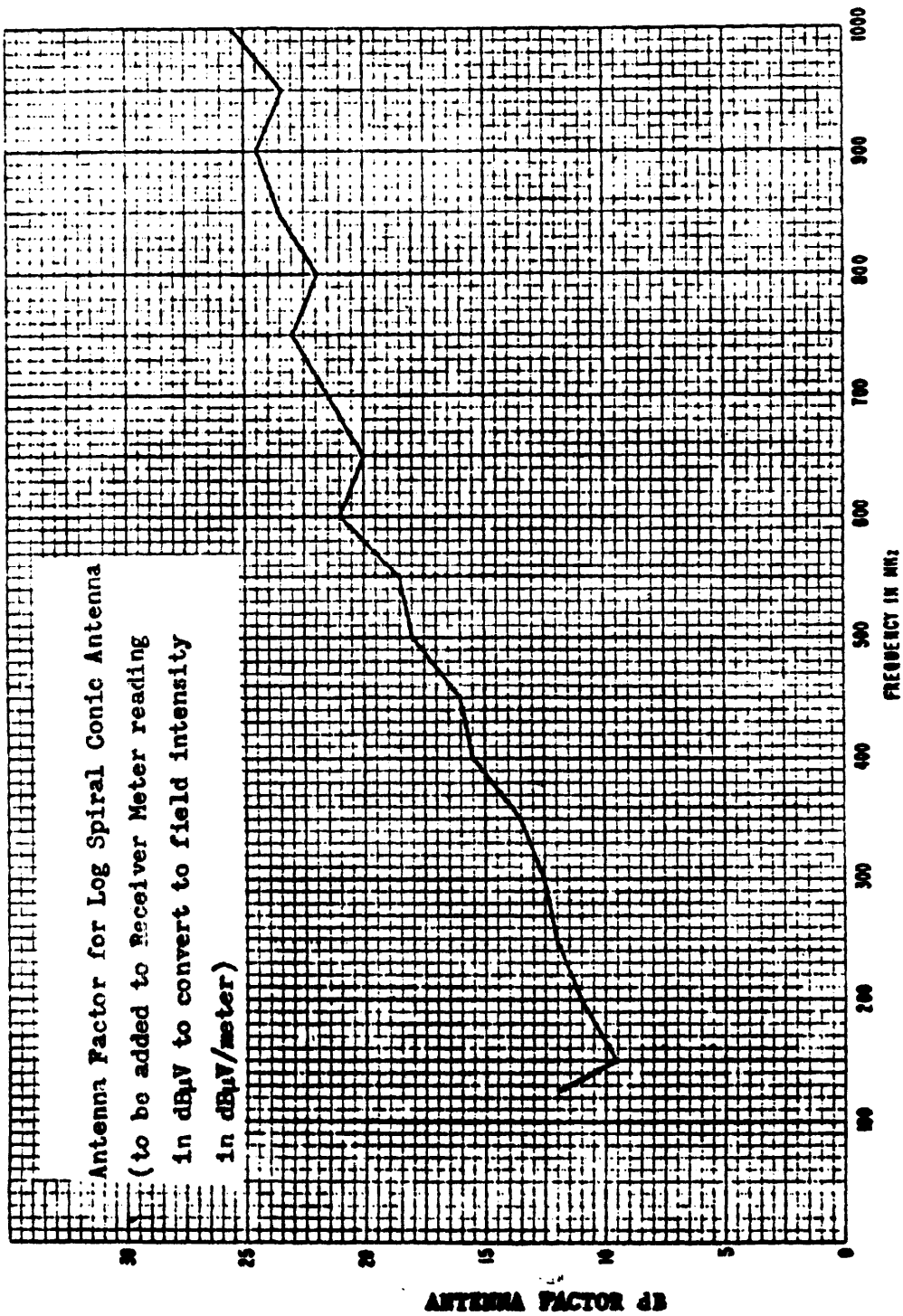


Figure 3

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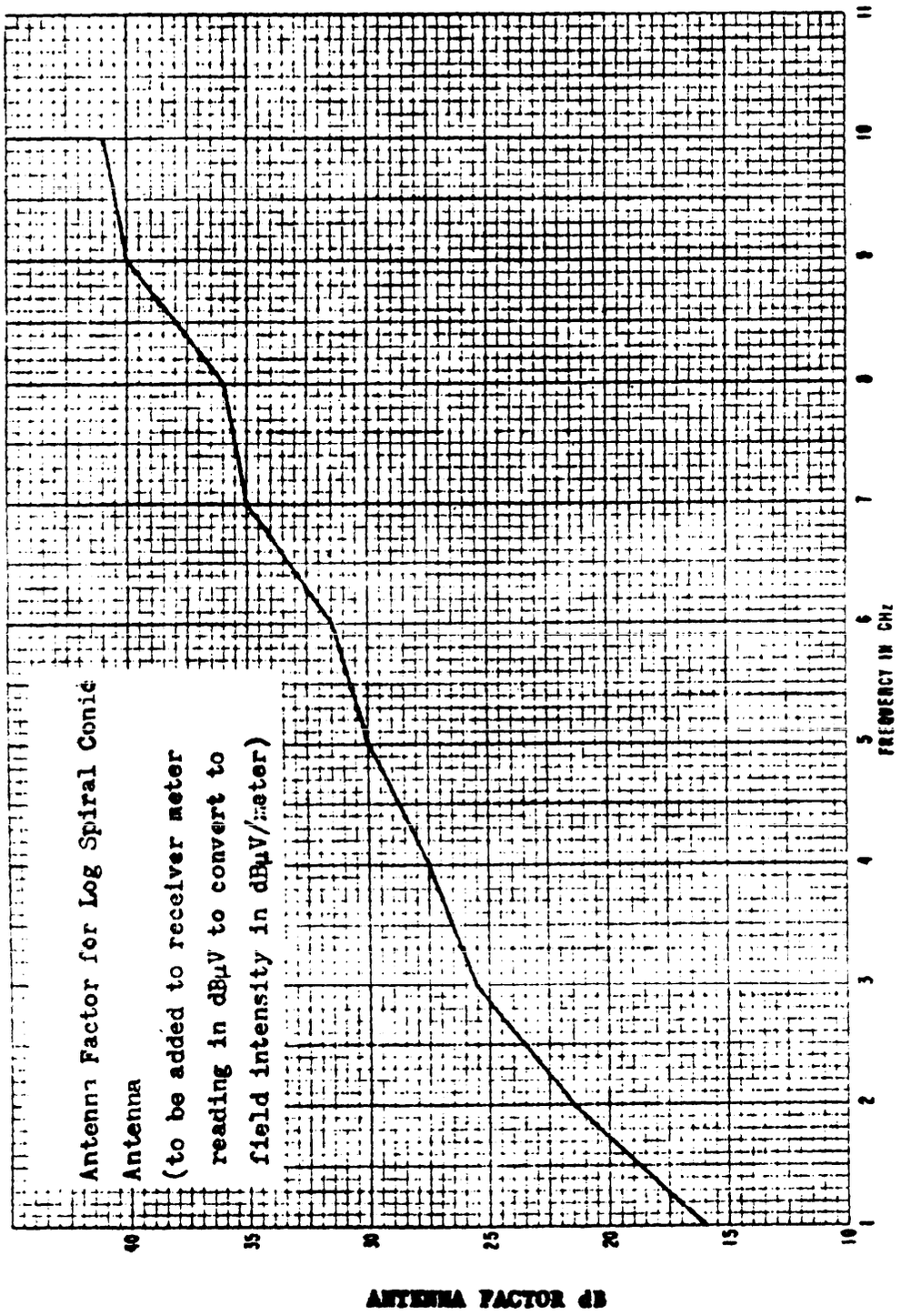


Figure 4

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4.10.2.2 Squelch overlap. - Under standard test conditions, with a 312.05 MHz input signal (amplitude modulated 30 percent at 1000 Hz), set the input signal level for a signal-plus-noise-to-noise ratio of 10 db measured at the receiver output. Reset the squelch control to just unsquelch the receiver. Vary the input signal level to determine that the receiver squelches and then unsquelches. Repeat the above test for the guard/EMER receiver at the guard frequency. These tests shall be performed to determine compliance with 3.17.2.2.

4.10.2.3 Squelch operation. - Under standard test conditions, with a 312.05 MHz input signal (amplitude modulated 30 percent at 1000 Hz), set the input signal for a signal-plus-noise-to-noise ratio of 10 db measured at the receiver output. Set the receiver output level to a reference of 50 mw. Reset the squelch control to just unsquelch the receiver and remove the input signal. Measure the receiver output. Without further squelch adjustment, repeat at one additional input signal frequency selected from each frequency group of 4.7.2. Repeat at the guard frequency for the guard/EMER receiver. These tests shall be performed to determine compliance with 3.17.2.3.

4.10.2.4 Squelch pulse response. - Under standard test conditions, apply a 312.05 MHz signal, pulse modulate the generator to provide a simulated RF signal with the required characteristics and verify compliance. These measurements shall be performed at the squelch sensitivity level (less than 4 microvolts for a 10 db signal-plus-noise-to-noise ratio with a 1000 Hz signal of 30 percent modulation) to determine compliance with 3.17.2.4.

4.10.3 IF selectivity. - Apply a 9.9 MHz $\pm$ 2.5 kHz CW signal to the IF input at a level sufficient to increase the agc line to a reference of 7.4 volts. Tune from 9.891 through 9.909 MHz and observe the ripple at the detector output to be less than 3.5 db. Increase the level by 6 db over the average passband reading and tune the signal generator frequency to the passband skirts for the same agc reference voltage. Increase the level another 54 db and tune the signal generator frequency for the same agc reference voltage. Increase the signal generator another 20 db and tune the signal generator frequency for the same agc reference voltage. Measure and record the signal generator frequency when the agc reference voltage is reached at each of the above levels. These tests shall be performed to determine compliance with 3.17.3.

4.10.4 Spurious response. - Apply a maximum of 4.0 uv RF input signal, modulated 30 percent at 1000 Hz, to produce a 10 db signal-plus-noise-to-noise ratio and 50 mw output. Increase the RF level by 60 db and scan the frequency region covering 200.00 to 425.00 MHz, excluding $\pm$ 200 kHz of the channel. Increase the RF level another 10 decibels and scan the frequency

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region beyond ± 10.0 MHz less ± 200 kHz of the channel frequency. Increase the RF level another 10 decibels and scan the frequency region beyond ± 10.0 MHz of the channel frequency. For any spurious response record the signal level required to produce a signal-plus-noise-to-noise ratio equal to or greater than 10 decibels and 50 mw output into a 150 ohm load to determine compliance with 3.17.4. Perform this test at five frequencies from 4.7.2 and at the guard frequency. When performing tests on the main receiver, the guard frequency shall be excluded as a test frequency. Tests performed on the guard receiver shall exclude the main receiver front panel dial frequency as a test frequency.

4.10.5 Blocking. - Perform the agc test of 4.10.9. Apply simultaneously to the receiver a 4.0 microvolt signal, modulated 30 percent at 1000 Hz, and a 1.0 volt signal unmodulated at the channel frequency. Remove the 1.0 volt signal and measure the recovery time to produce a 10 db signal-plus-noise-to-noise ratio. This test shall be performed at 399.50 MHz and the guard/EMER frequency to determine compliance with 3.17.5.

4.10.6 Noise rejection. - Apply an RF input test signal of 1000 microvolts, 90 percent amplitude modulated at 1000 Hz, at the receiver dial frequency. Adjust the audio for a 50 milliwatt output level. Introduce a second signal of the same carrier frequency pulsed on by a 10 ± 2 usec pulse with a repetition rate of 360 ± 60 Hz and a rise time of 1.0 usec maximum, and measure the resulting degradation in the 1000 Hz audio output with pulsed carrier amplitudes of 0.01 and 0.1 volts. Remove the modulation from the test signal, reduce the pulsed carrier amplitude to 1000 microvolts, and measure the resulting audio output. Measure the resulting audio output degradation when the carrier is pulsed on by a 2 ± 1 microsecond pulse with a pulse repetition rate of 900 ± 600 Hz and a rise time of 0.1 microsecond at carrier levels of 0.01 and 0.1 volts and the test signal is 1000 microvolts, 90 percent modulated at 1000 Hz. Remove the modulation from the test signal, reduce the pulsed carrier amplitude to 1000 microvolts and measure the resulting audio output. Perform this test at 227.50 234.75, 261.35, 295.40, 316.85, 342.60, 373.15 and 388.90 MHz, and the guard/EMER frequency to determine compliance with 3.17.6.

4.10.7 Desensitization. - Apply simultaneously to the receiver, a signal not greater than 4.0 microvolt to produce a 10 db signal-plus-noise-to-noise ratio, modulated 30 percent at 1000 Hz, on the channel frequency and a 60 db greater signal, modulated 30 percent at 400 Hz, at 1 MHz above and below the channel frequency. Measure the signal-plus-noise-to-noise ratio and perform this test at 254.50, 336.75 and 387.50 MHz and at the guard/EMER frequency to determine compliance with 3.17.7.

4.10.8 Quieting and internal noise. - Apply a 180 microvolt signal and measure the signal-plus-noise-to-noise ratio. Perform this test at

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eight frequencies, two from each group, selected from 4.7.2 and at the guard/EMER frequency to determine compliance with 3.17.8. (The guard/EMER receiver must be squelched while performing this test on the multichannel receiver. The multichannel receiver must be squelched while performing this test on the guard/EMER receiver.)

4.10.9 Automatic gain control. - Apply a standard test signal. Set the receiver output at 50 mw. Increase the input to 250 millivolts; except in the frequency range from 225.00 to 255.00 MHz, increase the input to 100 millivolts. Perform this test at four frequencies selected from 4.7.2 and at the guard frequency to determine compliance with 3.17.9.

4.10.10 Audio frequency response. - Apply a 180 microvolt signal, modulated 30 percent at 1000 Hz, to the receiver and set the audio output to 50 milliwatts. While holding the modulation percentage constant, adjust the audio frequency to 100, 300, 3000 and 12,000 Hz. Record the audio output. Perform this test at 268.05 MHz and the guard/EMER frequency to determine compliance with 3.17.10.

4.10.11 Audio adjust. - Apply a 275.50 MHz input signal of 180 microvolts modulated 30 percent at 1000 Hz, set the internal auxiliary audio control to full clockwise position and adjust the main audio control to the audio output levels specified. Set the main audio control to full clockwise position and adjust internal auxiliary audio control for 50 mw audio output. Set the main audio control to full counter-clockwise position and measure the audio output level. These measurements shall be performed to determine compliance with 3.17.11.

4.10.12 Audio distortion. - With an RF input of 250 millivolts, modulated 30 percent, except in the frequency range from 225 to 255 MHz with RF input of 100 millivolts, measure the distortion at audio frequencies of 300, 600, 1000 and 3000 Hz. Perform this test at 301.05 MHz, 374.30 MHz and at the guard frequency to determine compliance with 3.17.12.

4.10.13 Homing/DF aux. circuit. - Apply a 1000 microvolt signal, modulated 30 percent at 1000 Hz, to the receiver input. Observe the auxiliary output voltage. Vary the modulating frequency between 70 and 6000 Hz and observe the change in output voltage. Set the modulation to 30 percent at 200 Hz and compare input signal envelope and auxiliary output phases, using an oscilloscope. Perform the test at 255.55 MHz to determine compliance with 3.17.13.

4.10.14 Receiver self-test. - Depress and hold the receiver test button to verify the presence of the tone signal. Perform this test by first rotating the 10 MHz tuning knob through the entire range, then rotating the MHz knob, then rotating the kHz knob through the entire range. Release test button. These tests shall be performed to determine compliance with 3.17.14.

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4.10.15 Maximum RF input to receiver. - Operate the receiver under standard test conditions at 269.60 and 376.05 MHz. Apply a 10 volt open circuit peak signal at 269.50 and 376.05 MHz from a 50 ohm source for one minute to the receiver input. Remove the test signal and then verify specified receiver sensitivity (3.17.1) to determine compliance with 3.17.15.

4.10.16 Spurious quieting. - Under standard test conditions, with an RF input signal of 312.05 MHz amplitude modulated 30 percent at 1000 Hz, set the squelch for a 10 db signal-plus-noise-to-noise ratio measured at the receiver output. Remove the RF input signal. Each frequency group of 4.7.2 shall be randomly scanned until two channels are found where the squelch opens. These channels shall be those listed in 3.17.16. If a frequency is found where the squelch opens but it is not included in the list, all channels in that frequency group shall be scanned to assure that no more than one more unlisted channel exists. This test, in addition to the sensitivity test of 4.12 and the required sensitivity levels of 3.17.16, shall be made to determine compliance with 3.17.16.

4.10.17 Synthesizer operation. - With the signal generator set at the frequency of the dial setting of the radio set, the sensitivity test of 4.10.1 shall be met at each of the following frequencies 225.00, 231.15, 249.95, 250.10, 266.60, 277.70, 299.95, 300.10, 312.25, 322.20, 333.30, 344.40, 355.50, 366.60, 377.70, 388.85 and 399.95 to determine compliance with 3.17.17.

4.10.18 Audio output circuit. - With the receiver operating at 399.95 MHz and with 50 mw audio output, apply a short circuit to the receiver audio output for one minute. Remove the short circuit and operate the receiver into an open circuit for one minute. After the one minute operation, the unit shall meet sensitivity of 3.17.1 under standard test conditions to determine compliance with 3.17.18.

4.11 Electrical, transmitter. - All measurements are to be performed under standard test conditions unless otherwise noted and procedures are the same for the EMER transmitter.

4.11.1 Audio input circuits. - Place a decade resistance box in series with the audio input and measure the input impedance at 1000 Hz by adjusting the decade box for equal voltages (0.39 Vrms ± 10 percent) across the input and the decade box. Reverse the input terminals to demonstrate balance. Perform these tests to determine compliance with 3.18.1.

4.11.2 Power output, RF. - Measure the RF power output at 12 frequencies, three from each frequency group selected from 4.7.2. Reduce the supply voltage to 24.0 volts dc and measure the RF power output at 8 other frequencies, two from each frequency group selected from 4.7.2. These measurements shall be performed to determine compliance with 3.18.2.

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4.11.3 Modulation. - Operate the unit(s) in transmit mode at a modulation frequency of 1000 Hz. Measure the modulation percentage as determined through the use of the following relationship:

$$\text{Percent Modulation} = \frac{\text{E crest} - \text{E trough}}{\text{E crest} + \text{E trough}} \times 100$$

Perform this test at 225.00, 268.05, 312.05, 355.05 and 399.95 MHz to determine compliance with 3.18.3.

4.11.4 Frequency response, audio. - Place the unit(s) in transmit with 1000 Hz audio input adjusted to provide 30 percent modulation. Note the amplitude of the detected output. With the audio input level held constant, change the modulating frequency and measure the amplitude of the detected output. Perform the test at modulating frequencies of 50, 300, 3000, and 12,000 Hz and an RF frequency of 390.95 MHz to determine compliance with 3.18.4.

4.11.5 Modulation distortion. - Place the unit(s) in transmit with a zero dbm audio input. Measure the detected output with a distortion meter. Perform the test at 225.00, 268.05, 312.05, 355.05 and 399.95 MHz with modulation of 300, 600, 1000 and 3000 Hz to determine compliance with 3.18.5.

4.11.6 Carrier noise level. - Place the unit(s) in the transmit mode with no modulation. Note the detected audio output level. Apply an audio input of zero dbm at 1000 Hz and measure the increase in detected audio output. Perform the test at 274.70 MHz to determine compliance with 3.18.6.

4.11.7 Sidetone. - With a standard 50 ohm load, adjust the modulator assembly for a power level of 10 watts. Note sidetone presence. Reduce power level and determine sidetone presence at 8 watts. Further reduce power level until sidetone is absent. The power level shall be above 2.0 watts. Restore RF level to 10 watts. Connect mismatched loads which result in, first a 3:1 VSWR and sidetone presence, and second a 20:1 VSWR and sidetone absence. Operate the receiver under standard test conditions and use the audio output for a reference. Operate the transmitter with a sufficient 1000 Hz audio input for 30 percent modulation and measure the sidetone level. Performance of the audio tests of 4.10.10 and 4.10.12 adequately tests the frequency response and distortion of the sidetone circuits and previous performance of these tests constitutes performance of response and distortion measurements of the sidetone circuits. Perform tests at a frequency of 229.45 MHz to determine compliance with 3.18.7.

4.11.8 Continuous transmit. - To permit natural ventilation, suspend the unit(s) under test at least three (3) inches above the test bench and operate the transmitter continuously for one hour at 225.00 MHz and record

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the power output after 5, 20, 40 and 60 minutes. Receiver sensitivity measurements (4.10.1) shall be taken immediately before and after this test at 225.00 MHz and the guard/EMER frequency. Perform these tests to determine compliance with 3.18.8.

4.11.9 Retransmission. - Operate the receiver in the retransmit mode and observe the retransmit audio output and the signal presence dc switching action on the retransmit control (out) line when a 4.0 microvolt signal, with 30 percent modulation at 1000 Hz is applied. Operate the unit in the retransmit mode and observe that the transmitter is keyed on by a signal presence input signal on the retransmit control (in) line and that a 50 mw retransmit audio signal modulates the transmitter. Perform this test at 324.85 MHz and using a second radio set, measure retransmission distortion through the entire retransmission loop in both directions to determine compliance with 3.18.9.

4.11.10 Transmitter output circuit. - With normal RF power output conditions at 399.95 MHz apply a short circuit to the transmitter output for one minute. Remove the short circuit and operate the transmitter into an open circuit for one minute. After the one minute operation, the unit shall meet RF power output of 3.18.2 under standard test conditions to determine compliance with 3.18.10.

4.12 Frequency range. - Perform the sensitivity test of 4.10.1, the power output tests of 4.11.2 and the frequency accuracy test of 4.14 on all channel frequencies and guard/EMER receiver sensitivity and emergency power output and frequency accuracy to determine compliance with 3.19.

4.13 Frequency selection. - Operate the receiver under standard test conditions at 230.00 MHz and observe the audio output with an oscilloscope. Switch the three frequency controls (10 MHz, 1MHz, and 50 kHz step) to an adjacent position. Disable the squelch. Return the frequency control to 230.00 MHz. Measure the time required for normal audio (within 2 db of final output) to return. This test shall be performed to determine compliance with 3.20.

4.14 Frequency accuracy. - Without modulation and after the power has been off for a minimum of 5 minutes, apply input voltage and key the unit(s). Measure the RF output frequency at 225.00, 231.15, 249.95, 250.10, 266.60, 277.70, 299.95, 300.10, 312.25, 322.20, 330.30, 344.40, 355.50, 366.60, 377.70, 388.85, 399.95 MHz, and EMER, to determine compliance with 3.21.

4.15 Primary DC supply. -

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4.15.1 Input power. - The unit(s) shall be operated under standard test conditions in the receive and transmit modes. Measure the power input current in the receive mode and in the transmit mode. Perform this test at one (1) frequency, from each frequency group (See 4.7.2) to determine compliance with 3.22.1.

4.15.2 Normal voltage. - Operate the unit(s) under standard test conditions at input voltages of 24.0 Vdc and 29.0 Vdc. Measure RF power output (3.18.2), including EMER, receiver and guard/EMER sensitivity (3.17.1) and RF frequency (3.21) at one (1) frequency from each frequency group of 4.7.2. These tests shall be performed to determine compliance with 3.22.2.

4.15.3 Low voltage. - The supply voltage shall be adjusted to 21.0 Vdc and the power output (3.18.2) and sensitivity (3.17.1) of the unit(s) shall be measured to determine compliance with 3.22.3.

4.15.4 Emergency voltage. - The supply voltage shall be adjusted to 17 Vdc, the equipment turned on for one minute, with the transmitter keyed on for approximately thirty seconds during the one minute period. After restoring the supply to 27.5 Vdc, measure receiver sensitivity (3.17.1) and transmitter power output (3.18.2) at one frequency (See 4.7.2) to determine compliance with 3.22.4.

4.15.5 Power source transient. - Under standard test conditions, except apply an 80 volt pulse for an 0.2 second duration to the power input, apply each pulse (plus and minus) three (3) times within one (1) minute as specified in MIL-STD-704 Category "B". These tests shall be performed to determine compliance with 3.22.5.

4.15.6 Power supply protection. - Turn the equipment on and determine that the radio set is operational. Ground (short) the + 5 Vdc line at a convenient test point and obtain an inoperable condition. Remove the short circuit to determine compliance with 3.22.6. Repeat this test on the + 9 Vdc, + 15 dc, and + 18 Vdc lines.

4.16 EMER Transmit & Receive. - Set the dial frequency to a standard test frequency (See 4.7.2) and the function switch to EMER position. Measure the emergency transmit frequency and RF power output. The frequency shall be 243.00 Mhz ± 8 khz and the power output shall be not less than 8 watts. Measure the sensitivity of the guard/EMER receiver and the main receiver. The sensitivity of the guard/EMER receiver shall be not more than 4.0 uv for a 10 db signal-plus-noise-to-noise ratio and the main receiver shall be disabled. This test shall be performed to determine compliance with 3.23.

4.17 X-Mode. -

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4.17.1 Input, wideband. - The wideband input impedance shall be measured at 10 kHz and the impedance shall be greater than 100 ohms. Using a square wave signal of 20 kHz at 12 volts peak to peak, clamped to ground, applied to the transmitter audio input through either a 100 ohm or 780 ohm source impedance. Measure, using either an oscilloscope or digital modulation meter, the percentage of modulation and the plain cipher control line to determine compliance with 3.24.1.

4.17.2 Output, wideband. - With a 1000 Picofarad load capacitor in parallel with a 50 k ohm resistor applied to the receiver wideband output, while receiving the 20 kHz square wave modulated 70 to 90 percent, measure the wideband output to determine compliance with 3.24.2.

4.17.3 Output, narrowband. - The narrowband output from the receiver shall be terminated in 600 ohms in parallel with 150 ohms and the received 1 kHz reference tone level modulated 30 percent shall be measured with and without the plain cipher control signal to determine compliance with 3.24.3.

4.17.4 Output, guard channel. - Terminate the guard channel only output with 600 ohms and measure the received 1 kHz reference tone level modulated 30 percent. When not receiving digital information, verify on the guard channel that the signal is received via the narrowband output and measure the guard channel only output to determine compliance with 3.24.4.

4.17.5 Output, receiver. - A complete transmitter-receiver system shall be set up and it shall be determined that receiving of plain traffic with a cipher control input is possible. In addition, the operation of the guard only channel with 15 kHz square wave being received on the main channel shall be observed. These tests shall be made using a 1 kHz tone to simulate the plain traffic on the main channel and guard traffic on the guard channel only output. Any control signals required for cipher mode operation of the receiver shall be applied. The 1 kHz tone shall be transmitted on the frequency of the main receiver. Application or removal of the plain cipher mode control signal shall result in less than 3 db degradation of the received 1 kHz tone. Simultaneous reception of the 15 kHz square wave input on the main channel and a 1 kHz tone on the guard channel shall result in less than 3 db degradation to the received 1 kHz tone on the guard channel only output as the 15 kHz square wave signal is applied or removed. The tests and measurements shall be made to determine compliance with 3.24.5.

4.17.6 Operational, X-Mode. - The unit(s) shall be connected with a TSEC/KY-28 and operated in receive and transmit modes on a minimum of six (6) randomly selected channels. Monitor the received and transmitted signal for normal communication capability to determine compliance with 3.24.6.

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4.18 Visual and mechanical. - Prior to assembly into the unit(s). printed circuit boards shall be examined for compliance with MIL-P-55110 and the equipment drawings. Assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-252. All external moveable controls, access covers and the test button shall be operated mechanically. Assembled units(s) shall be examined for defects listed in MIL-STD-252. This inspection shall be performed to determine compliance with 3.30.

4.19 Interchangeability. - The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the unit(s). Mechanical dimensions and electrical performance required by the gage and equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.27. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured prior to assembly into the units:

a. Mechanical (dimensions). - The gage points covered by the drawings listed on the applicable gage lists specified in the solicitation/ contract.

b. Electrical performance (Items). - Items shall be tested to determine compliance with the performance requirements specified on the applicable gage lists specified in the solicitation/contract.

4.20 Operational inspection. - The final adjustments of the unit(s) shall be verified. Each unit(s) shall be operated in the receive and transmit modes on four (4) frequencies from each frequency group as specified in 4.7.2 and on the emergency frequency in the "EMER" position, and shall be operated in the receive mode on 243.00 MHz in the "T/R GUARD" mode with the dial frequency set at a frequency other than 243.00 MHz. This inspection shall be performed to determine compliance with 3.28.

4.21 Weight. - The unit(s) shall be weighed to determine compliance with 3.5.

4.22 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance with the packaging requirements of Section 5.

4.23 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the operational inspection test 4.20 shall be conducted to determine freedom from operational malfunction caused by rough handling.

DEPARTMENT OF DEFENSE SINGLE
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4.24 Turn Around Time. -

4.24.1 Turn Around Time, transmit to receive. - Under standard test conditions, with a 268.05 MHz input signal amplitude modulated 30 percent at 1000 Hz, set the receiver squelch for a signal-plus-noise-to-noise ratio of 10 db measured at the receiver output. Increase the RF input level to 50 uv. Use the transmitter key line voltage to trigger an appropriate storage oscilloscope. While monitoring the receiver audio output on the vertical axis of the oscilloscope, key and unkey the transmitter while switching from the receiver RF input signal to the transmitter RF output signal using a suitable coaxial relay and load. The length of time required from when the transmitter is unkeyed until full receive output occurs shall be measured to determine compliance with 3.25.1.

4.24.2 Turn Around Time, receive to transmit. - Under standard test conditions, simultaneously apply an unmodulated input signal and the attenuated output of the transmitter at 268.05 MHz, 70 percent modulation at 1000 Hz, to the inputs of an appropriate mixer. The output of the mixer shall be applied to the vertical axis of an appropriate storage oscilloscope. Note the level at the crest of the modulated signal on the oscilloscope. Use the transmitter key line voltage to trigger the oscilloscope. While monitoring the mixer output on the vertical axis of the oscilloscope, key the transmitter. Measure the length of time required from when the transmitter is keyed until the modulation crest level previously noted occurs to determine compliance with 3.25.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Radio Set AN/ARC-116( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Radio Set AN/ARC-116( ) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Each Radio Set AN/ARC-116( ) shall be packaged in accordance with items 1 through 8 of Table VI and Figure 5.

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BILL OF MATERIAL
TABLE VI

EF No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Literature	1	IC-1	10-1/2	8	1/4							
	Bag	1		12	9		MIL-B-117	II	C				
2	Radio Set AN/ARC-116	)1	IIb	8	5-3/4	4-7/8	(Bare wt. 8.6 lbs.)						
3	Case, Top	1		12	8-1/2	4-3/16	MIL-P-19644 (2 lbs. per cu. ft. density)	N					
4	Desiccant	2		5	3-1/2	5/8	MIL-D-3464 (4 unit)	II					
5	Case, Bottom	1		12	8-1/2	4-3/16	MIL-P-19644 (2 lbs. per cu. ft. density)	N					
6	Humidity Indicator	1		3	2		MIL-I-8835						
7	Bag, Barrier	1		25	18		MIL-B-131		2				
8	Box, Fiberboard	1		12-3/8	9	8-5/8	PPP-B-636	CF	WR	SW	W5c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

AMSEL-MR/P Form 3135
1 Oct 1965

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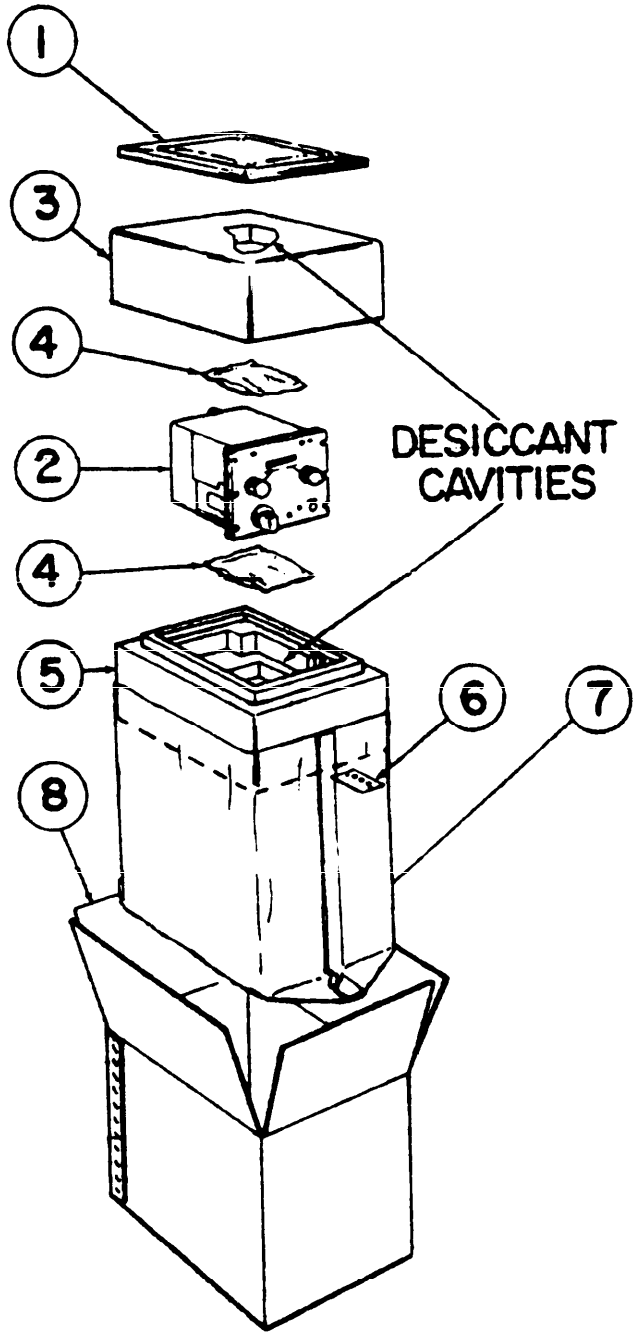


Figure 5

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5.1.2 Level C. - Radio Set AN/ARC-116( ) shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2).

5.2.1 Level A. -

5.2.1.1 Consolidation. - A quantity of Radio Set AN/ARC-116( ), packaged as specified in 5.1 shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, Class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load Type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, Type IV, Size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, Class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed with tape conforming to PPP-T-76 in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length or width is 48 X 24 inches or more and the weight exceeds 100 pounds, 3 X 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type 1, Class B.

5.2.2 Level B. - (See 6.6)

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5.2.2.1 Consolidation. - A quantity of Radio Set AN/ARC-116( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers, packed as specified in 5.2.2.1, shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, Type IV, as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of Radio Set AN/ARC-116( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1, shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The intended use of the Radio Set AN/ARC-116( ), is receiving and transmitting clear-voice or X-Mode communication when installed in aircraft.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation and packaging, and packing. (See Section 5).

c. When rough handling and functional tests are required.

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d. First Article inspection:

Several samples of each item cited in Section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while electrical performance measurements can be made on other sample units. (See 3.3)

e. Marking and shipping of sample.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of spare parts required.

i. Submission of the statement of treatment referenced in 3.14 should be specified in the solicitation/contract.

6.3 Group C inspection. - Approval to ship unit(s) on order may be withheld, at the discretion of the Government, pending decision from the procuring activity, until compliance with 4.5.4 is complete.

6.4 Location of operational inspection. - It is desirable that the operational inspection 4.20 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected at the packaging site immediately prior to packaging.

6.5 Nomenclature. - The parentheses in the nomenclature shall be deleted or replaced by a letter identifying the particular design, for example AN/ARC-116W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Deviation. - Request for deviation in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed designed change to withstand the applicable test for the selected method, including the rough handling test in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing material selected and economies realized.

6.7 Dimensional data. - Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the equipment cited in the Table. When dimensions of the items vary from those cited, the dimensions of the packaging materials shall be adjusted accordingly.

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6.8 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractors quality control system and the quality history of the product.

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Preparing Activity:
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Project No. 5821-A109

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-5566A(EL) RADIO SET, AN/ARC-116( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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